



Loan Default Prediction using Machine Learning and Explainable AI (SHAP and LIME).

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Masters Thesis for the partial fulfilment
of the requirements for the Master Degree
in Msc Information Technology with Data Analytics

University of the West of Scotland
School of Computing
27 April, 2026

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DEDICATIONS

We would like to express our sincere gratitude to our supervisor, **Dr. Yan Ge**, for his guidance, support and valuable feedback throughout this project. His advice helped us improve the quality of our work and stay focused during the research process.

We would also like to thank our module lead, **Dr. Jamal Hwidi**, for his support and guidance throughout the dissertation module and **Dr. Volkan Tunali**, programme leader, for the academic support provided during our studies. We also appreciate the valuable role of **Dr. Fadi Safieddine** as a moderator.

We also acknowledge the support and academic environment provided by **University of the West of Scotland (London Campus)** during the completion of this project and the course.

As a group, we would like to acknowledge the collaborative effort of all four team members in contributing to this project. Finally, we would like to thank our families and friends for their encouragement and support throughout this journey.

ABSTRACT

This study aims to develop a reliable and interpretable system for predicting loan default using machine learning and explainable artificial intelligence (XAI). The main objective is to compare simple baseline models, such as Logistic Regression and Decision Tree, with advanced models like Random Forest and XGBoost and to understand how well they predict loan default.

The dataset is first prepared using data cleaning, handling missing values, encoding, and balancing techniques such as SMOTE. Different models are then trained and tested using performance measures including accuracy, precision, recall, and F1-score. In addition, explainability techniques, SHAP and LIME, are applied to understand how the models make predictions and which features are most important.

The results show that advanced models, especially XGBoost, perform better than baseline models in predicting loan default. However, they are more complex and harder to understand. The use of SHAP and LIME helps explain these complex models by showing how different features, such as interest rate and credit-related factors, affect the prediction.

Overall, the study achieves its goal by creating a balanced framework that provides both high prediction accuracy and clear explanations. This can help financial institutions make better and more transparent lending decisions.

Keywords: Loan Default Prediction; Credit Risk Assessment; Machine Learning; Explainable Artificial Intelligence; SHAP; LIME; XGBoost; SMOTE; F1-Score

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Chapter 1: Introduction

In recent years, financial institutions have increasingly used machine learning models to improve the accuracy of credit risk assessment. This project aims to develop an Explainable Artificial Intelligence (XAI) framework for predicting loan default using large-scale financial data. The study begins with traditional baseline models as benchmarks and then moves to more advanced ensemble models to improve predictive performance. At the same time, the framework focuses on ensuring that credit decisions are not only accurate, but also transparent and justifiable.

1.1 Background and Rationale

The accurate prediction of loan defaults is a fundamental requirement for financial stability, allowing institutions to mitigate credit risk and optimize lending strategies. Traditionally, this assessment relied on "white-box" statistical models such as Logistic Regression and Decision Trees, which remain the industry standard because they provide the transparency necessary to satisfy regulatory requirements and justify credit decisions (Ajayi et al., 2021). However, as financial datasets grow in complexity, these traditional methods often struggle to capture the intricate, non-linear relationships that influence borrower behaviour.

To improve predictive accuracy, the financial sector has shifted toward advanced ensemble techniques like Random Forest and XGBoost. While these models offer superior performance, they function as "black boxes," making it difficult to explain the logic behind a loan denial—a significant challenge in a regulated environment that demands fairness and accountability (Ribeiro et al., 2016). This project is justified by the need to establish reliable machine learning models while leveraging XAI techniques to provide a clear understanding of how borrower characteristics influence default risk.

1.2 Problem Statement

The increasing use of advanced machine learning models in credit risk prediction has significantly improved predictive accuracy. However, it has also introduced challenges related to transparency and interpretability. Explainable Artificial Intelligence (XAI) methods, such as LIME introduced by Ribeiro et al. (2016), have demonstrated the importance of generating interpretable explanations to support trust in automated decision-making. More recent studies have shown that SHAP provides more consistent and reliable feature attributions compared to

LIME, particularly in financial applications such as credit default prediction (Gramegna and Giudici, 2021).

Despite these advancements, a critical limitation persists. Existing research indicates that XAI techniques can produce unstable and inconsistent explanations when applied to real-world datasets characterised by class imbalance, a common issue in credit risk modelling where default cases are relatively rare (Chen et al., 2024). Furthermore, most studies evaluate interpretability methods independently, without systematically integrating baseline models, ensemble techniques and XAI methods within a unified framework. This creates a gap between model performance and trustworthy interpretability, limiting the practical adoption of such systems in regulated financial environments.

Therefore, there is a need to develop a robust and integrated approach that combines high-accuracy machine learning models with stable and reliable XAI techniques, ensuring both predictive performance and consistent interpretability in credit risk prediction.

1.3 Aim

The aim of this project is to develop and evaluate an accurate and interpretable credit risk prediction framework by comparing baseline and ensemble machine learning models, and by integrating explainable AI techniques to provide reliable and transparent predictions on imbalanced financial data.

1.4 Research Questions

RQ1: How do baseline models compare with ensemble models in predicting loan default risk?

RQ2: How does class imbalance affect the performance of machine learning models in credit risk prediction?

RQ3: How effective are XAI techniques such as SHAP and LIME in explaining model predictions?

RQ4: How consistent are the explanations provided by SHAP and LIME on the dataset?

RQ5: What is the trade-off between model accuracy and interpretability?

RQ6: Can a framework be developed that balances predictive performance with explainability for credit risk prediction?

1.5 Objectives

- To implement and evaluate baseline models such as Logistic Regression and Decision Tree.
- To develop and compare ensemble models such as Random Forest and XGBoost.
- To apply explainable AI techniques (SHAP and LIME) to interpret model predictions.
- To analyse the trade-off between model accuracy and interpretability.
- To identify key features influencing loan default.
- To propose an interpretable and reliable credit risk prediction framework.

1.6 Project Scope

The scope of this research is structured into four interrelated components:

- **Baseline Modelling:** Developing Logistic Regression and Decision Tree models to serve as interpretable benchmarks.
- **Ensemble Learning:** Designing advanced models like Random Forest and Gradient Boosting to capture complex data patterns.
- **Explainability Application:** Applying SHAP and LIME to interpret complex models and generate global and local explanations.
- **Comparative Analysis:** Conducting a systematic empirical comparison of SHAP and LIME, assessing their stability and suitability in the context of financial regulations.

Chapter 2: Literature Review

This chapter reviews the existing literature related to machine learning and explainable artificial intelligence (XAI) in credit default prediction. It aims to provide a theoretical and empirical foundation for understanding how machine learning models can be used to predict credit risk while maintaining interpretability.

The chapter first presents the theoretical review, covering key concepts such as credit risk theory, machine learning models and XAI techniques like LIME and SHAP. This is followed by the empirical review, which examines previous studies on credit default prediction and the use of explainability methods. Finally, the chapter identifies research gaps and presents the conceptual framework for the study.

2.1 Theoretical and Conceptual Review

This study is grounded in key theoretical perspectives that explain how machine learning models can achieve both predictive accuracy and interpretability in financial decision-making. These theories help to understand the relationship between prediction performance, transparency and trust in credit risk models.

2.1.1 Credit Risk Theory

Credit risk theory informs how borrower characteristics are linked to default probability and provides the basis for predictive modelling in lending. Financial institutions use this theory to assess the probability of default (PD) and support lending decisions. Traditionally, this has been carried out using statistical models such as logistic regression, which analyse borrower characteristics including income, credit history, and repayment behaviour.

The theory highlights that accurate prediction of default risk is essential to reduce defaults and maintain stability in the financial system. However, traditional statistical methods seem to perform with lesser efficiency while capturing complex and non-linear relations in large datasets thereby limiting their predictive performance. This limitation has led to the increasing use of machine learning models in credit risk assessment (Nwafor and Nwafor, 2023; Hlongwane et al., 2024).

2.1.2 Accuracy–Interpretability Trade-off Concept

One of the key concepts in this area is the accuracy–interpretability trade-off, which suggests that there is a balance between how accurate a model is and how easy it is to understand.

Simple models, such as logistic regression, are highly interpretable because their results can be easily explained. However, they may not always achieve the highest prediction accuracy. In contrast, advanced models such as neural networks, XGBoost and random forest generally produce more accurate predictions, but these models are considered to have lack of transparency and so referred to as “black-box” models.

This trade-off creates a challenge in credit default prediction, where financial institutions require both high predictive performance and clear explanations of model decisions. Therefore, this concept focuses on the need for approaches such as Explainable Artificial Intelligence (XAI), with the aim of balancing both accuracy and interpretability (Hlongwane et al., 2024; Chen et al., 2024).

2.1.3 Black-Box Models Concept

The concept of black-box models refers to complex and tedious methods whose internal working is not understood by humans. These models take inputs and generate outputs, but the reasoning behind their predictions is not transparent.

In credit default prediction, black-box models can achieve high predictive accuracy, but they raise concerns related to trust, fairness, and accountability. For example, when a loan application is rejected, banks are expected to give justification on their decisions. Without transparency, stakeholders may find it difficult to trust the model.

This concept highlights the need for explainability techniques that can make complex models more understandable and interpretable (Chen et al., 2024; Gramegna and Giudici, 2021).

2.1.4 Explainable Artificial Intelligence (XAI) Framework

Explainable Artificial Intelligence (XAI) focuses on improving the transparency and interpretability of machine learning models. Rather than replacing complex models, XAI provides methods to explain how predictions are made.

In credit default prediction, XAI helps financial institutions identify which factors influence a borrower's risk level. It also improves trust in model outputs and ensures that decisions can be justified to both customers and regulators.

Explainable AI is crucial in connecting advanced predictive models with real-world requirements such as transparency, fairness, and accountability (Gramegna and Giudici, 2021; Ribeiro et al., 2016; Chen et al., 2024). This is particularly relevant to this study, which focuses on improving interpretability through baseline models and feature analysis.

Explainable AI methods can be mainly of two types namely global and local which explains overall model behaviour and local predictions respectively. Both are important in credit risk modelling. Global explanations help to know the factors influencing loan default, while local explanations can support understanding of decisions made for specific borrowers (Ribeiro, Singh and Guestrin, 2016). However, although XAI improves interpretability, some studies suggest that explanations may vary depending on the model and dataset used, which remains a challenge for reliable implementation (Chen, Calabrese and Martín-Barragán, 2024). In addition, SHAP and LIME offer different strengths, with SHAP often providing broader feature-level insights, while LIME focuses on explaining individual predictions (Gramegna and Giudici, 2021).

2.1.5 Shapley Value (SHAP) Concept

Shapley values are based on game theory principles and estimate the contribution of each feature to a prediction. Here, features act as contributors, while the model output represents the outcome.

SHAP (SHapley Additive exPlanations) applies this concept to explain model predictions by providing every variable a contribution score according to its impact on the final output. For example, it can show how factors of a borrower like loan amount, interest rate, loan types, etc. influence the probability of default.

This approach is widely used in credit risk modelling because it provides a consistent and mathematically sound way to interpret predictions. It also supports both global explanations of model behaviour and local explanations of individual predictions (Gramegna and Giudici, 2021; Chen et al., 2024).

2.1.6 Local Explanation (LIME) Concept

The concept of local explanation is considered as a concept that even if a model is complex overall, its behaviour can be explained locally for a specific prediction. One of the most widely used techniques based on this concept is LIME (Local Interpretable Model-Agnostic Explanations).

LIME explains a prediction by using a simpler model to copy how the main model behaves around one specific case. This helps show why a borrower is predicted as either high risk or low risk.

This method is particularly useful in credit default prediction, where financial institutions often need to explain individual decisions rather than the overall model behaviour. Local explanations help improve transparency and support better decision-making (Ribeiro et al., 2016; Gramegna and Giudici, 2021).

2.1.7 Conclusion of Theoretical Review

In summary, the theoretical foundation of XAI in credit default prediction is built on several key ideas, including credit risk theory, the accuracy-interpretability trade-off, black-box model concept and XAI models. These theories, concepts and ideas explain why there is a need to balance predictive accuracy with transparency in financial decision-making.

By applying XAI, financial institutions can improve model performance while still providing clear and understandable explanations. This helps to increase trust, ensure regulatory compliance and support better decision-making in credit risk management. However, despite these developments, challenges remain in balancing interpretability and predictive performance, which highlights the need for further research.

2.2 Empirical Review

2.2.1 Empirical studies on traditional models in credit risk

Traditional models such as logistic regression and decision trees have been widely applied in credit scoring because they are simple and interpretable. Hlongwane et al. noted that logistic regression remains one of the most common credit risk models in practice due to its simplicity and ability to produce understandable predictions, even though it is often outperformed by more advanced models in terms of accuracy (Hlongwane *et al.*, 2024). Similarly, Serengil et

al. (2022) compared logistic regression with several other machine learning methods for non-performing loan prediction and found that logistic regression was useful as a benchmark model, although more advanced algorithms achieved stronger overall performance.

These findings show that traditional models still have value in financial applications, especially where transparency is important. However, their limitation is that they cannot properly describe the complex and nonlinear relationships which reduces their predictive strength in large and diverse datasets.

2.2.2 Empirical studies on machine learning models in credit risk

A lot of empirical studies suggests that advanced methods of machine learning are better performing than traditional methods in loan default prediction. Yang et al. (2025) evaluated logistic regression, decision tree, random forest, XGBoost, LightGBM and other models on the Home Credit dataset and found that gradient boosting methods achieved the best overall results, with XGBoost slightly outperforming the other models in accuracy, precision and recall. In another study, Akinjole et al. (2024) compared decision tree, random forest, SVM, XGBoost, AdaBoost and multilayer perceptron on LendingClub data and reported that ensemble methods produced very strong predictive results, especially when class imbalance was addressed properly.

Nwafor and Nwafor (2023) also found that ensemble learning techniques and deep learning methods can identify determinants of non-performing loans more effectively than conventional approaches because they handle complex nonlinear relationships in large datasets better. Overall, the empirical evidence suggests that advanced machine learning models improve prediction accuracy, but this often comes at the cost of lower interpretability.

2.2.3 Empirical studies on XAI in credit risk prediction

As machine learning models became more complex, empirical studies began to focus on explainability. Ribeiro et al. (2016) introduced LIME as a model-agnostic explanation technique and showed that it can help users understand individual predictions of black-box classifiers by approximating them locally with an interpretable model. In the credit risk domain, Gramegna and Giudici (2021) applied both SHAP and LIME to SME credit risk data and examined their usefulness in explaining complex predictions, showing that explainability methods can support post-hoc feature interpretation in financial applications.

Chen et al. (2024) further explored interpretability in imbalanced credit scoring datasets and found that the stability of SHAP and LIME explanations decreases as class imbalance increases, which suggests that explainability itself can be affected by data characteristics. Hlongwane et al. (2024) also demonstrated that Shapley values can be used to build interpretable credit scorecards from advanced tree-based models, helping bridge the gap between predictive performance and practical transparency.

These empirical studies show that XAI methods are increasingly important in credit risk modelling because they improve trust, transparency and regulatory acceptability.

2.2.4 Summary of empirical findings

The empirical literature shows three main patterns. First, baseline models like logistic regression and random forest remain important because they are easy to interpret and useful as baseline models. Second, ensemble models such as XGBoost and random forest generally achieve better predictive performance. Third, explainable methods giving shapley values and local explanations are becoming essential for explaining the behaviour of complex models in credit risk settings.

The literature shows a progression from traditional interpretable models to advanced ensemble methods, and then towards explainable AI approaches that address transparency concerns. This suggests predictive performance and interpretability should not be treated as separate goals, but as interconnected requirements in credit risk modelling. These themes together provide the foundation for this study.

Although many studies compare model performance, fewer studies examine baseline models in depth together with feature analysis. This suggests a need for research that not only builds interpretable baseline models but also studies the importance of individual features in loan default prediction, which is the main aim of this study.

2.3 Research Gap

Despite the significant progress in credit risk prediction, several limitations remain in the existing literature. Many studies have focused primarily on improving predictive accuracy by using advanced machine learning models such as Random Forest, XGBoost, and deep learning techniques. While these techniques often achieve strong performance, they are commonly treated as complex methods where it is intricate to make predictions

At the same time, traditional baseline models are relevant because of their interpretability. However, in many studies, these models are used only as baseline comparisons and are not analysed in detail. This limits the understanding of how interpretable models perform and how they are utilized effectively in crucial financial decision.

In addition, although some suggestions have been made regarding use of XAI techniques such as LIME and SHAP to improve transparency, these methods are often applied only to complex models. There is limited research that integrates baseline models, ensemble models, and explainability techniques within a single framework to provide both accurate predictions and meaningful interpretations.

Furthermore, feature analysis is often treated as a secondary aspect of model development. Many studies focus on overall model performance without thoroughly examining which variables have the greatest influence on loan default prediction. This creates a gap in understanding the key factors driving credit risk.

Therefore, there is a need for a structured approach that combines baseline models, ensemble models, model evaluation and feature analysis with explainability techniques. This report is targeted to fill the gap by introducing and comparing interpretable baseline models and advanced ensemble models, while also analysing feature importance and applying XAI methods to improve transparency in loan default prediction.

Chapter 3: Methodology

This section explains how the project is carried out to achieve the research objectives. The report focuses on developing machine learning models for predicting loan default and to analyse the importance of different features influencing the prediction. This design ensures that the process is logical, systematic and suitable for solving the research problems.

Machine learning techniques are used because they are effective in analysing large-scale financial data and finding patterns that may be harder to identify using traditional methods. In addition, the project focuses on explainability by analysing feature importance, which helps in understanding how the models make decisions.

3.1 Research Approach

Since the study predicts loan default using numerical financial data and machine learning techniques, it is based on a quantitative research strategy. The issue is handled as a supervised binary classification challenge, in which models anticipate default or non-default outcomes by identifying patterns in labelled data. Additionally, a quantitative approach makes it possible to assess model performance using metrics like recall, accuracy and precision.

3.2 Problem Formulation

The loan default prediction task is formulated as a binary classification problem, where the model predicts whether a borrower is likely to default or not. The input consists of borrower-related variables like income, loan types, interest rates, etc. which are used to identify patterns associated with default behaviour. This formulation allows usage of standard predictive models and evaluation metrics that support credit risk assessment.

3.3 Conceptual Framework

The conceptual framework outlines the process from data input to prediction and explanation.

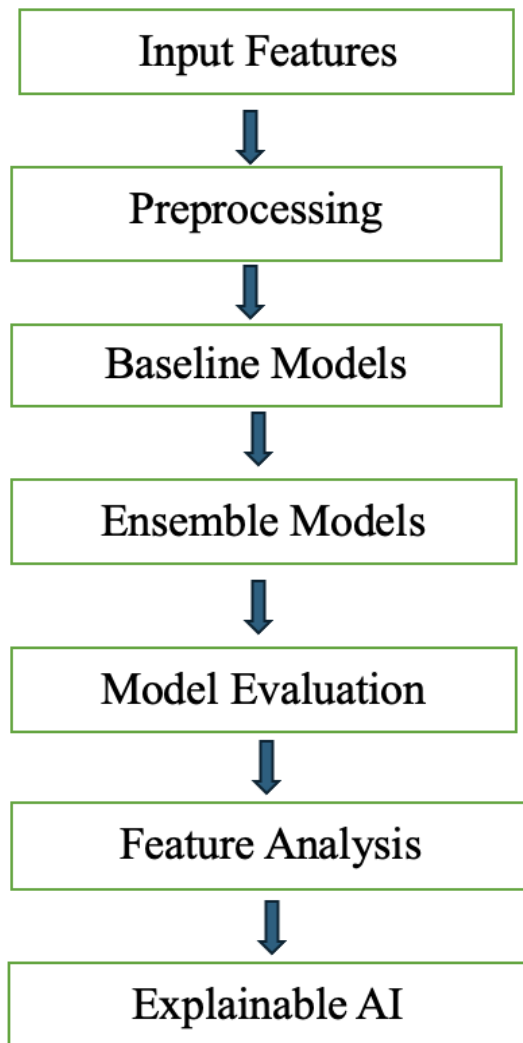


Figure 1: Conceptual Framework of the Study

Source: Authors

The conceptual framework illustrates the overall process of loan default prediction while maintaining interpretability. It shows how borrower-related features are processed and used to train baseline models along with advanced machine learning models.

The models are evaluated using evaluation criteria or metrics, along with feature evaluation to know the key influencing variables. XAI methods such as SHAP and LIME, are then applied to give clear explanations of model predictions.

Overall, the framework integrates prediction and interpretability to support transparent and reliable decision-making in credit risk assessment.

3.4 Data Source and Description

The study uses a publicly available secondary dataset from Kaggle ('Loan Default Dataset'), containing anonymised financial and demographic borrower information. The dataset includes both numerical and categorical features, along with a labelled target variable indicating default or non-default. Initial exploration was carried out to understand the data structure and support model development.

3.5 Data Preprocessing

Data preprocessing was carried out to prepare the dataset for machine learning modelling. This included removing irrelevant features, handling missing values, encoding categorical variables, splitting the data, addressing class imbalance using SMOTE, and applying feature scaling. These steps were necessary to improve data quality, reduce bias and ensure the dataset was suitable for both baseline and ensemble models.

3.5.1 Removal of Irrelevant Features

Non-informative features such as ID and year were removed because they had no predictive value. Features with high missing values, including upfront charges and related variables, were also removed. Interest rate spread was excluded due to redundancy with rate of interest.

3.5.2 Handling Missing Values

After removing irrelevant features, the next step was to handle missing values in the dataset. Missing data can affect model performance if not treated properly, so appropriate imputation methods were used based on the type of variables.

Missing values were handled using mode imputation for categorical variables and median imputation for numerical variables. Median was chosen to reduce the influence of outliers in financial data.

3.5.3 Encoding of Categorical Variables

After handling missing values, categorical variables were converted into a numerical format so that they could be used in machine learning models. Since most algorithms cannot process text-based data directly, encoding is an essential step in preprocessing.

One-hot encoding was applied to convert categorical variables into numerical form. One category from each variable was dropped to reduce multicollinearity, particularly for Logistic Regression.

Although one-hot encoding increases the number of features, it allows the models to better capture relationships within the data and ensures that categorical information is effectively used in the prediction process.

3.5.4 Data Splitting Strategy

After preprocessing and encoding, the dataset was divided into training, validation, and test sets to evaluate model performance effectively. A 70:15:15 split was used, where 70% of the data was allocated for training, 15% for validation and 15% for testing.

The training set was used for model development, while hyperparameter tuning was conducted using 5-fold cross-validation within the training data through GridSearchCV. The validation set was used as an additional performance check to confirm model stability before final evaluation on the independent test set.

Stratified sampling was applied during the splitting process to maintain the proportion of default and non-default cases across all subsets. This ensures that each dataset represents the original distribution and helps produce more reliable and unbiased results.

3.5.5 Handling Class Imbalance

In classification problems such as loan default prediction, the dataset is often imbalanced, and one class has significantly more observations than the other. To address this issue, the Synthetic Minority Over-sampling Technique (SMOTE) was applied. SMOTE works by generating synthetic samples of the minority class, helping to balance the dataset and improve the model's ability to learn patterns related to loan default.

SMOTE was applied only to the training data to address class imbalance and improve the detection of default cases. This helped improve recall and F1-score while avoiding data leakage.

3.5.6 Feature Scaling

After handling class imbalance, feature scaling was applied to ensure that numerical variables are on a similar scale. This is important because some machine learning models, such as Logistic Regression, are sensitive to differences in the scale of input features.

In this study, standardisation was used to scale the numerical features. This method transforms the data so that it has a mean of zero and a standard deviation of one. As a result, all numerical variables contribute equally to the model without being dominated by features with larger values.

It is important to note that feature scaling was applied only to the training data and then the same transformation was applied to the validation and test sets.

3.6 Model Selection

In this project, both baseline and advanced machine learning models are used to predict loan default. This approach allows for a comprehensive evaluation of model performance and helps identify the most suitable model for credit risk prediction.

The baseline models selected are Logistic Regression and Decision Tree. These models are simple, widely used and easy to interpret. Logistic Regression is suitable for binary classification problems and provides clear insights into how each feature influences the probability of default. However, as a linear model, it may struggle to capture complex non-linear relationships in financial data.

Decision Tree can capture non-linear relationships and presents results in a rule-based structure, making it easy to understand. However, Decision Trees can be prone to overfitting and may not generalise as well as more advanced models when used alone.

In addition to baseline models, ensemble models such as Random Forest and XGBoost are also implemented. These models are more advanced and can handle complex patterns in the data, often resulting in improved predictive performance. Random Forest helps reduce overfitting by combining multiple trees, while XGBoost is effective in optimising performance through boosting techniques. However, these models can be less interpretable than simpler baseline models.

By combining both baseline and ensemble models, the study can compare interpretability and accuracy. Baseline models provide transparency and ease of explanation, while ensemble models focus on achieving higher performance. This integrated approach supports a more balanced evaluation of loan default prediction and reflects the trade-off between predictive accuracy and interpretability.

3.7 Model Evaluation Metrics

To evaluate model performance, multiple evaluation metrics are used, as relying on a single metric may not provide a complete assessment.

Accuracy measures the overall correctness of the model, but it may not be reliable for imbalanced datasets. Precision evaluates how many predicted default cases are correct, while recall measures how well the model identifies actual default cases. Recall is particularly important in credit risk, as missing a default case can have serious consequences. Similarly, F1-score provides a balance between precision and recall, making it useful when both false positives and false negatives need to be considered.

In addition, ROC-AUC is used to measure the model's ability to distinguish between default and non-default cases across different thresholds. A higher AUC indicates better performance.

A confusion matrix is also used to give a detailed breakdown of prediction results, helping to understand the strengths and weaknesses of the model.

3.8 Explainable Artificial Intelligence (XAI) Methods

In addition to model development, this study also incorporates Explainable Artificial Intelligence (XAI) techniques to improve the interpretability of model predictions. In financial applications such as credit risk assessment, it is important not only to make accurate predictions but also to understand how these predictions are made.

Two widely used XAI methods, SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations) are applied in this study.

SHAP is used to measure the contribution of each feature to the model's predictions. It provides both global explanations, showing overall feature importance, and local explanations, explaining individual predictions. This helps identify the key factors influencing loan default.

LIME is used to explain individual predictions by approximating complex models with simpler interpretable models. It provides local explanations for specific instances, helping to understand why a particular borrower is classified as high risk or low risk.

By using SHAP and LIME, the study improves transparency and allows better interpretation of both baseline and ensemble models. This supports more reliable and explainable decision-making in credit risk prediction.

3.9 Feature Analysis Methods

Feature analysis is used to understand which variables have the most influence on loan default prediction. Correlation analysis is used to check relationships between variables and identify any strong associations. In Logistic Regression, feature importance is analysed using coefficients, which show how each variable affects the probability of default.

For tree-based models such as Decision Tree, Random Forest, and XGBoost, feature importance is based on how much each feature contributes to splitting the data.

In addition, SHAP and LIME are used to explain model predictions. These methods help understand how features influence both overall results and individual predictions.

3.10 Tools and Technologies

This project is implemented using Python. Pandas and NumPy are used for data processing, and Matplotlib and Seaborn are used for visualisation. Scikit-learn is used to build baseline models such as Logistic Regression and Decision Tree. XGBoost is used for advanced models. SHAP and LIME are used for model explanation. Other additional tools were used to facilitate the procedure.

3.11 Ethical Considerations

This study uses a publicly available dataset that does not contain any personal or sensitive information. There is no involvement of human participants, so no ethical approval is required.

To ensure fairness and transparency, interpretable models and explainability methods such as SHAP and LIME are used to provide clear explanations of predictions.

Chapter 4: Implementation, Testing and Results

This chapter presents the practical part of the project, including data preparation, model implementation, testing, and the presentation of results. It explains how the models were developed, how their performance was evaluated, and how explainability methods were used to interpret the predictions.

4.1 Dataset Overview

The dataset used in this study consists of 148,670 records with multiple financial and demographic features related to borrowers. These features include variables such as income, loan amount, property value, credit score, and loan characteristics.

The dataset contains both numerical and categorical variables, providing a comprehensive view of borrower profiles. The target variable, *Status*, is binary, where 0 represents non-default and 1 represents default.

This dataset provides a suitable basis for applying machine learning models to predict loan default and analyse the key factors influencing credit risk.

4.2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the structure and characteristics of the dataset before further steps.

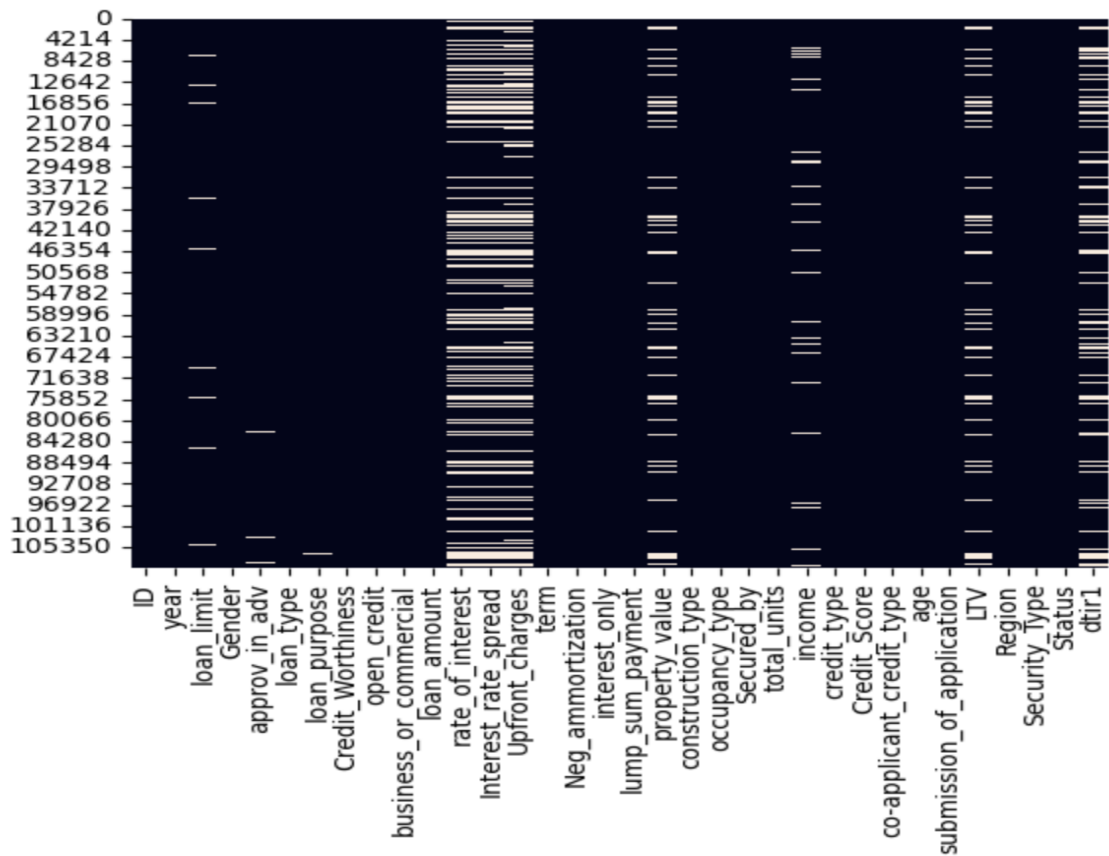


Figure 2: Heatmap Showing Missing Values
Source: Authors

Figure 2 shows the distribution of missing values across the dataset. It can be observed that some features contain a significant number of missing values, particularly interest-related variables. This supports the need for appropriate preprocessing techniques such as removal or imputation.

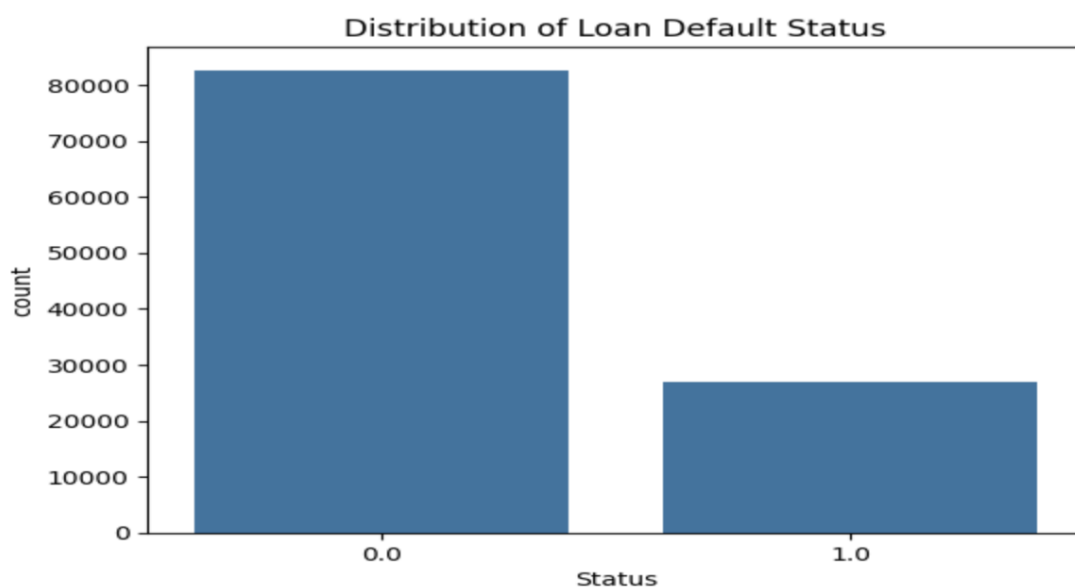


Figure 3: Bar Plot showing Distribution of Target Variable
Source: Authors

Figure 3 presents the distribution of the target variable (*Status*). The dataset is imbalanced, with a higher number of non-default cases compared to default cases. This highlights the need for balancing techniques such as SMOTE during model training.

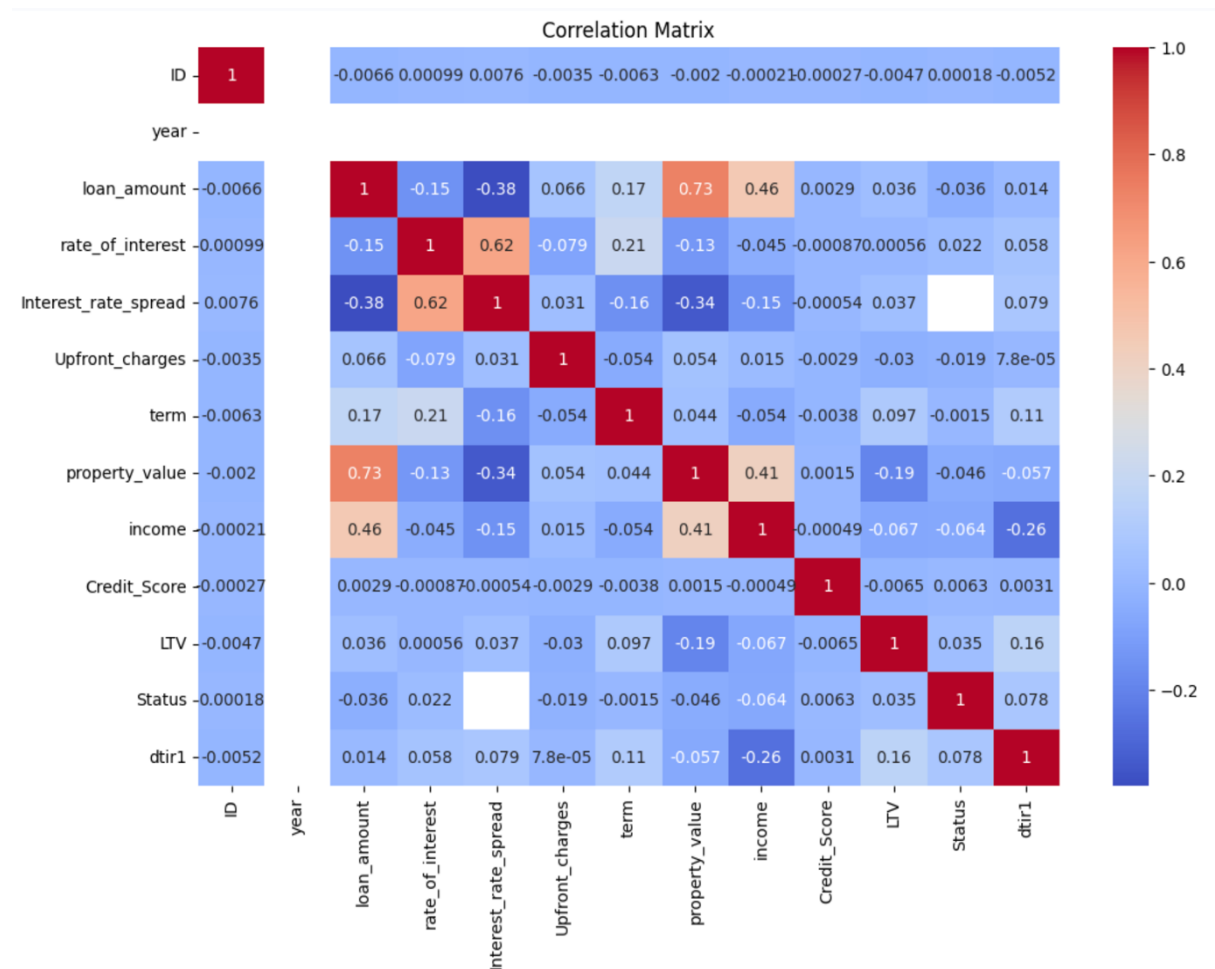


Figure 4: Correlation Matrix of Numerical Features
Source: Authors

Figure 4 shows the correlation between numerical features. Most variables have weak relationships with the target variable, indicating that loan default prediction involves complex patterns. However, strong correlations are observed between some features, such as loan amount and property value, suggesting possible redundancy.

Overall, the EDA provides important insights into the dataset and supports the selection of appropriate preprocessing and modelling techniques.

4.3 Data Preprocessing

The preprocessing steps described above in Chapter 3 were implemented prior to model training. Irrelevant and redundant features were removed, missing values were imputed, and

categorical variables were encoded using one-hot encoding. The dataset was then split into training, validation and test sets using a 70:15:15 ratio.

To address class imbalance, SMOTE was applied to the training data, followed by feature scaling using standardisation. These steps prepared the dataset for model development and evaluation.

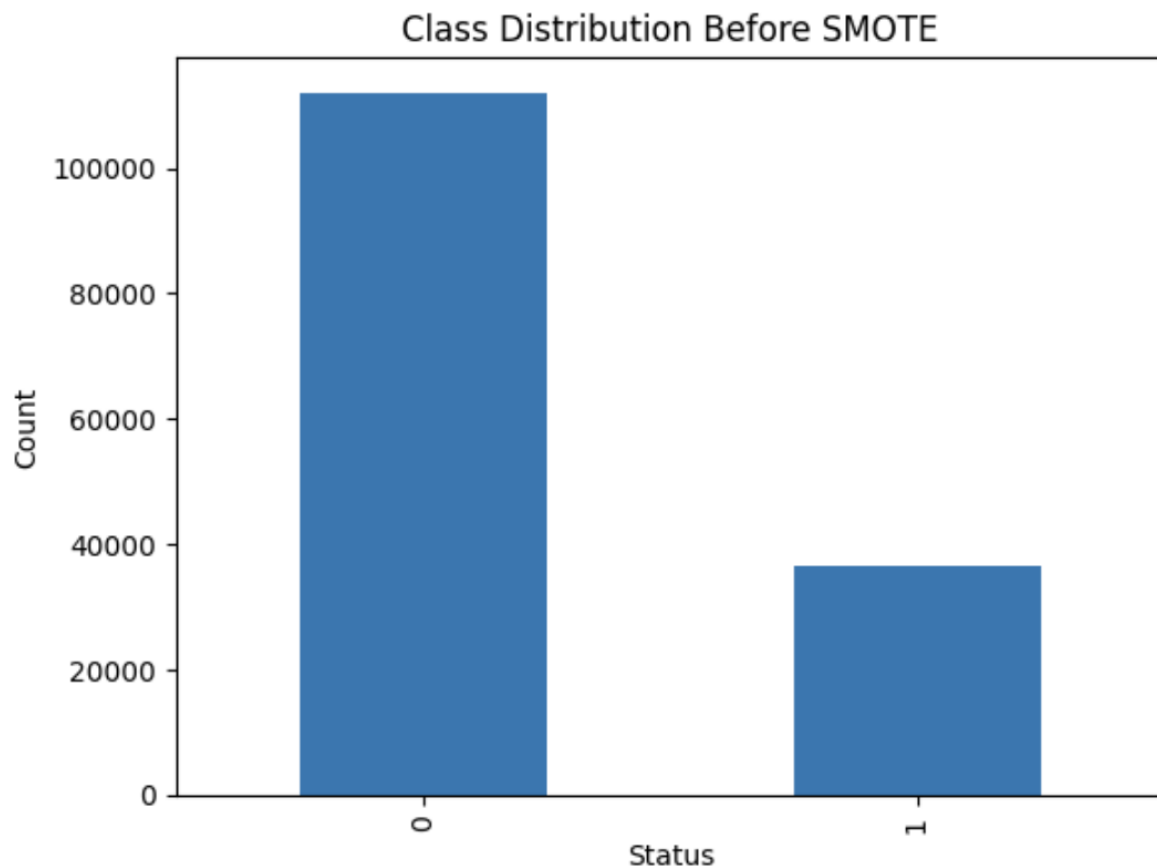


Figure 5: Class Distribution before SMOTE

Source: Authors

The dataset is imbalanced, with a higher number of non-default cases compared to default cases.

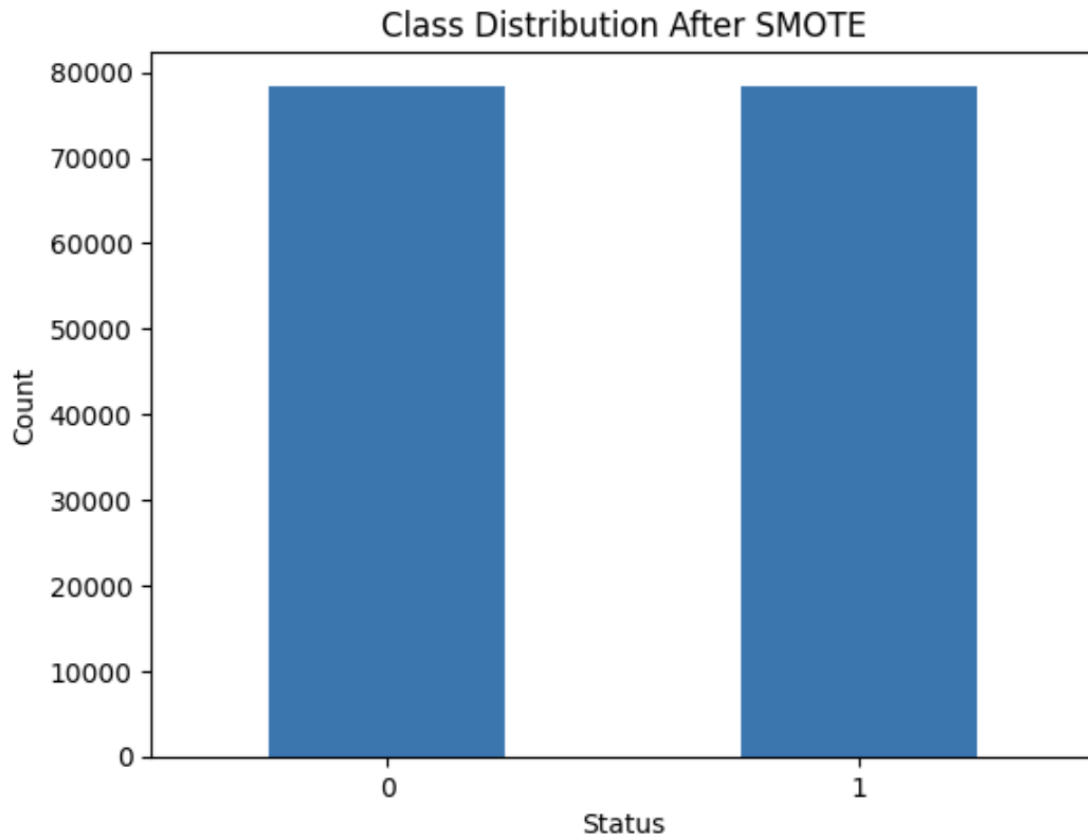


Figure 6: Class Distribution after SMOTE
Source: Authors

After applying SMOTE, the dataset becomes balanced, with equal representation of both classes.

4.4 Model Implementation

In this study, multiple machine learning models were implemented to predict loan default, including both baseline and ensemble approaches.

4.4.1 Logistic Regression

Logistic regression was used as a baseline model due to its simplicity and interpretability. This model is appropriate for estimating the probability of loan default

4.4.2 Decision Tree

The Decision Tree model was implemented to capture non-linear relationships in the data. It uses a rule-based structure to split the data into different branches, making it easy to interpret.

4.4.3 Random Forest

One of the Ensemble learning methods used was Random Forest to improve prediction performance. It combines multiple decision trees to reduce overfitting and improve generalisation.

4.4.4 XGBoost

XGBoost was also implemented as an advanced ensemble model as it is known for its high performance and ability to handle complex patterns. It uses boosting techniques to improve prediction accuracy by focusing on difficult cases.

For model training, Logistic Regression was applied to scaled data, while the tree-based models were trained using the resampled dataset. Hyperparameter tuning was performed for Random Forest and XGBoost using GridSearchCV with 5-fold cross-validation on the training data. In addition, the validation set was used as an additional performance check to assess model stability before final evaluation. After this, all models were evaluated on the independent test set using multiple performance metrics.

4.4.5 Validation Performance Check

Before final testing, the separate validation set was used to assess the stability of model performance after tuning. The validation results showed similar performance between Random Forest and XGBoost, indicating consistent model behaviour. Random Forest achieved a validation F1-score of 0.864, while XGBoost achieved 0.861. However, XGBoost showed higher validation recall of 0.902. Although Random Forest achieved a marginally higher validation F1-score, XGBoost produced higher recall, which is important in detecting default cases. These results supported the robustness of the selected models prior to final testing.

4.5 Model Testing and Results

The performance of the implemented models was evaluated using multiple metrics, including accuracy, precision, recall, F1-score, AUC-ROC and PR-AUC. These metrics provide a comprehensive assessment of model performance, especially in the presence of class imbalance.

Table 1: Performance Comparison of Machine Learning Models

Model	Accuracy	Precision	Recall	F1-score	AUC-ROC	PR-AUC
Logistic Regression	0.839	0.732	0.548	0.627	0.798	0.723
Decision Tree	0.913	0.818	0.834	0.826	0.886	0.723
Random Forest	0.928	0.837	0.881	0.858	0.983	0.949
XGBoost	0.927	0.821	0.901	0.859	0.983	0.951

Source: Authors

Table 1 presents the performance of all models using multiple evaluation metrics. It can be observed that ensemble models outperform the baseline models across most metrics. XGBoost achieved the highest F1-score (0.859) and recall (0.901), indicating its strong ability to correctly identify default cases. Random Forest also performed well, achieving the highest accuracy (0.928) and a high AUC-ROC (0.983), showing strong overall classification performance.

The Decision Tree model showed moderate performance, with balanced precision and recall, while Logistic Regression recorded the lowest performance, particularly in recall (0.548) and F1-score (0.627). This indicates that Logistic Regression struggles to detect default cases effectively.

Overall, XGBoost and Random Forest demonstrate superior performance due to their ability to capture complex patterns in the data, while baseline models provide lower predictive accuracy but better interpretability. Although Random Forest achieved slightly higher accuracy, XGBoost was selected as the final model due to its stronger recall, highest F1-score, and superior AUC-ROC and PR-AUC performance, making it more suitable for credit default prediction.

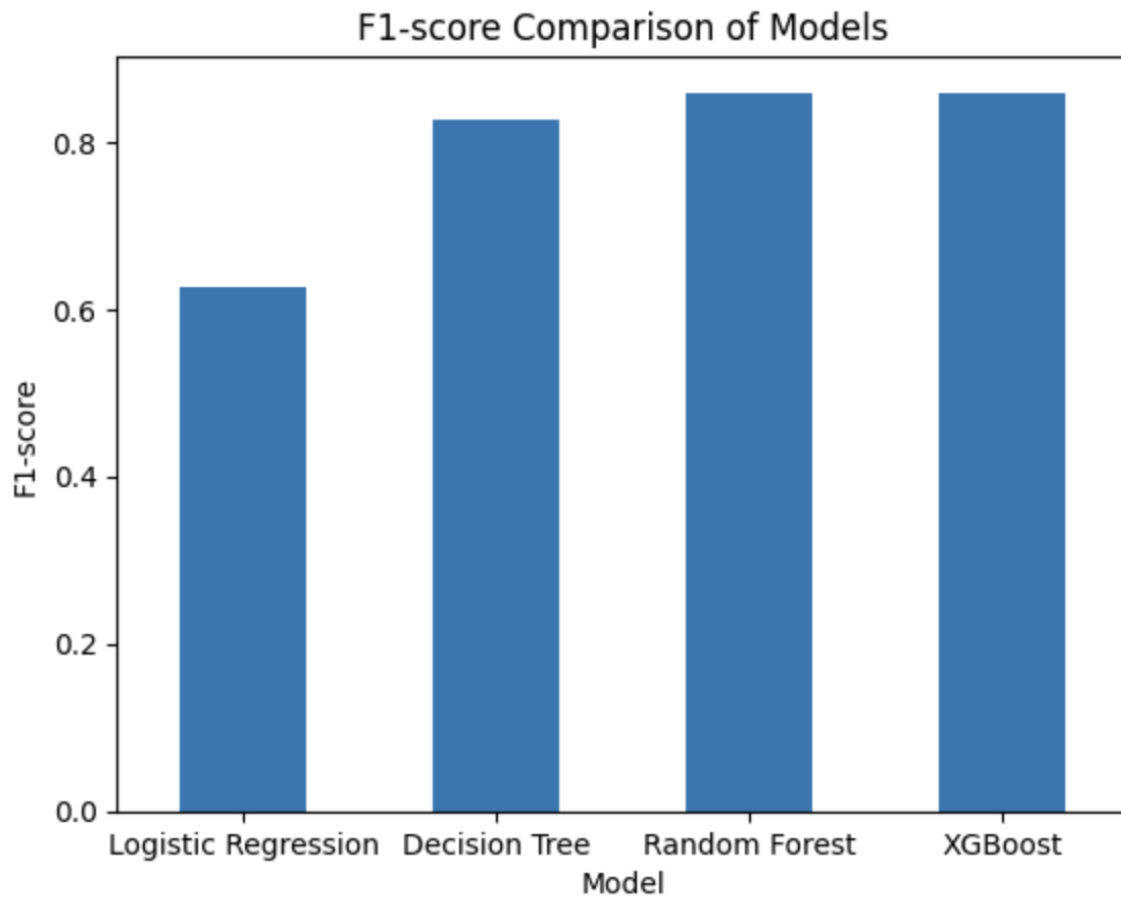


Figure 7: Comparison of models based on F1-Score
Source: Authors

Based on these results, XGBoost was selected as the best-performing model for further analysis.

4.6 Explainability Analysis

To improve the interpretability of the model, Explainable Artificial Intelligence (XAI) techniques were applied. The best-performing model, XGBoost, was selected for explanation based on its performance in the previous section.

SHAP

SHAP (SHapley Additive exPlanations) was used to provide global interpretability by identifying the most important features influencing the model's predictions. The SHAP summary plot was generated to highlight the contribution of each feature. Features with higher SHAP values have a stronger impact on the model's output.

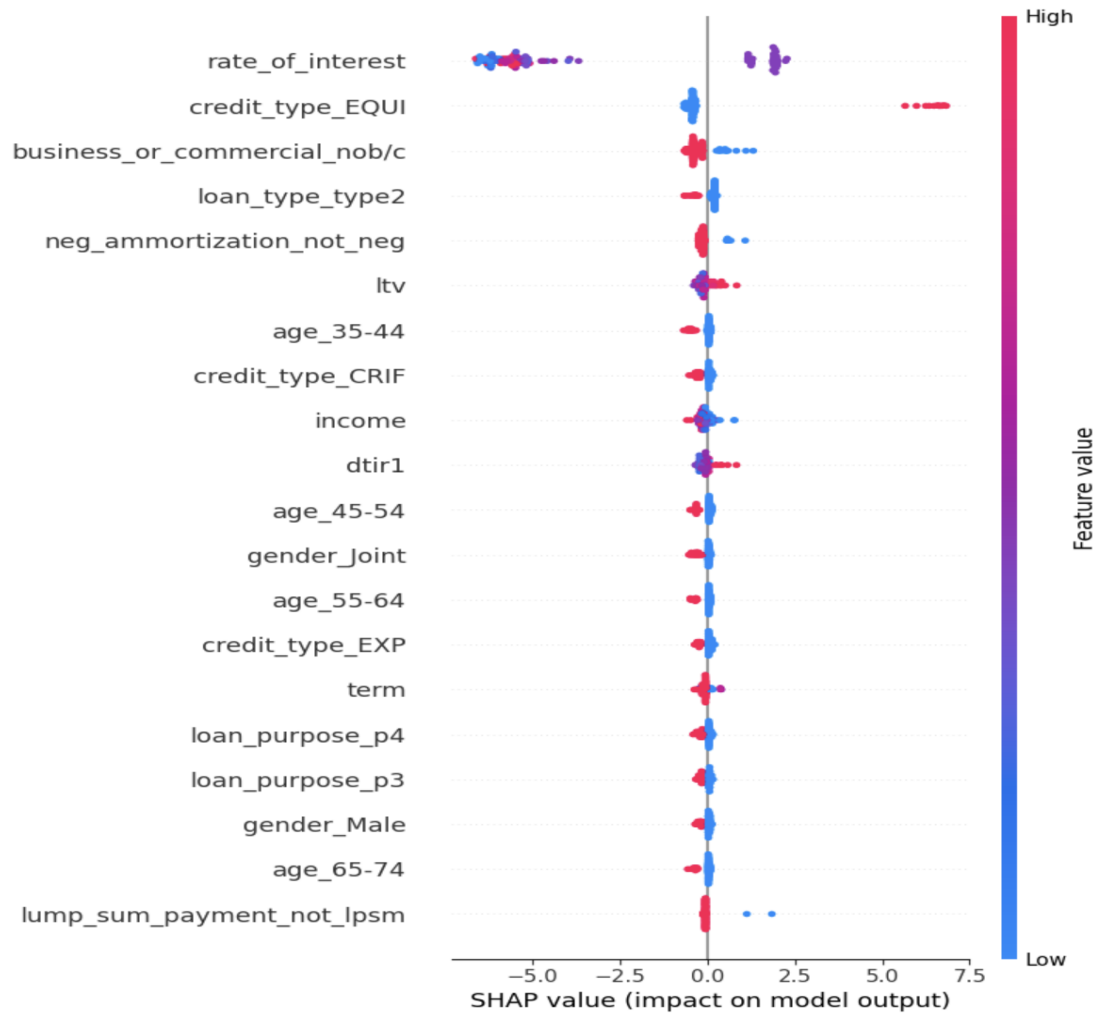


Figure 8: SHAP Summary Plot showing Feature Importance
Source: Authors

Figure 8 shows the SHAP summary plot for the XGBoost model. The plot ranks features based on their importance in predicting loan default. Features at the top have a greater influence on the model's predictions.

Figure 8 shows the SHAP summary plot for the XGBoost model. The plot ranks features based on their importance in predicting loan default. Features at the top have a greater influence on the model's predictions.

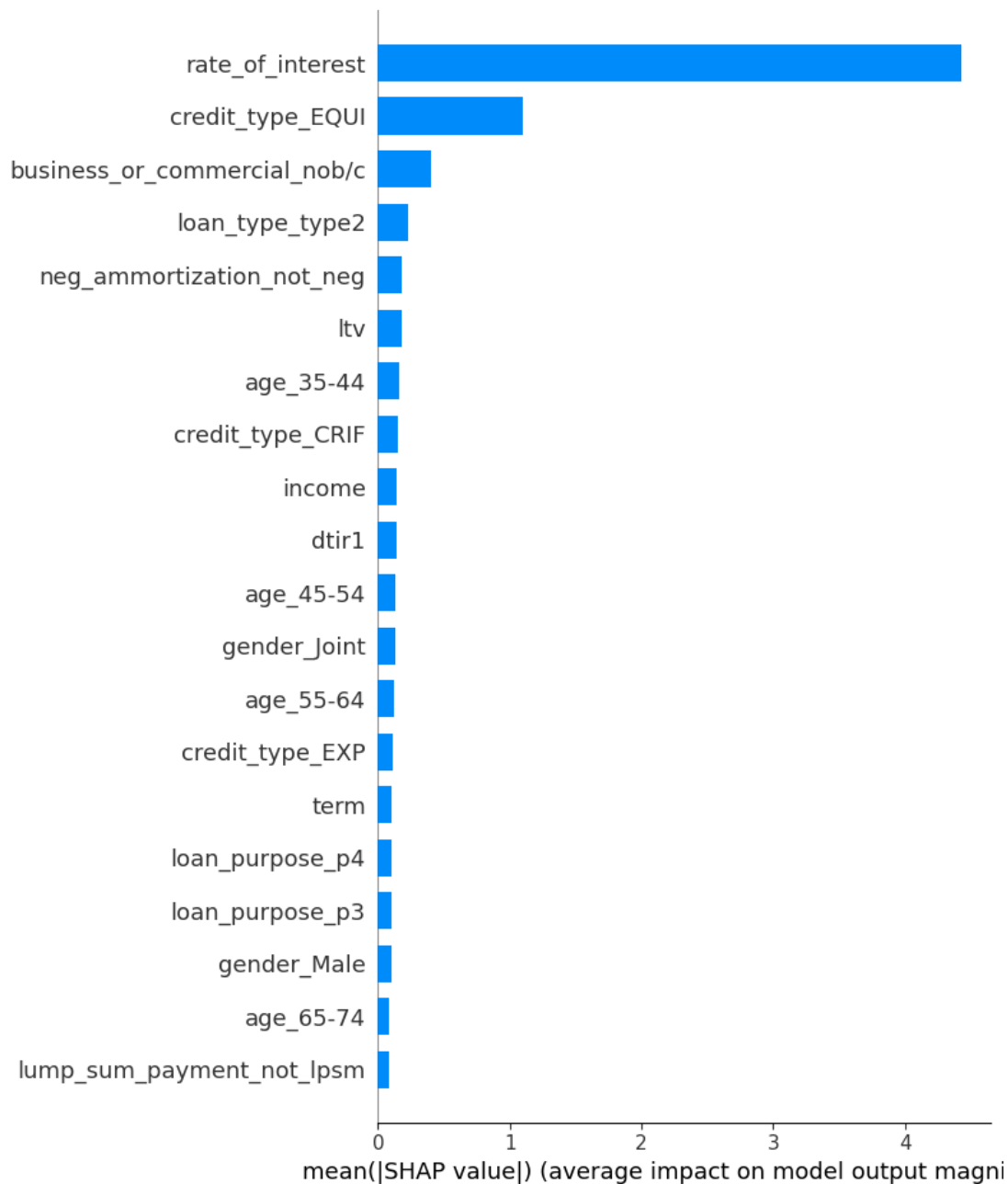


Figure 9 Mean Absolute SHAP Values (Feature Importance Ranking)
Source: Authors

The bar plot displays the average impact of each feature on the model's output. **rate_of_interest** emerges as the most influential feature by a significant margin, followed by **credit_type_EQUI**. Other notable features include business or commercial loan type, loan type, negative amortization status, and LTV (Loan-to-Value) ratio. This ranking aligns well with domain knowledge in credit risk, where interest rate serves as a strong proxy for borrower risk profile.

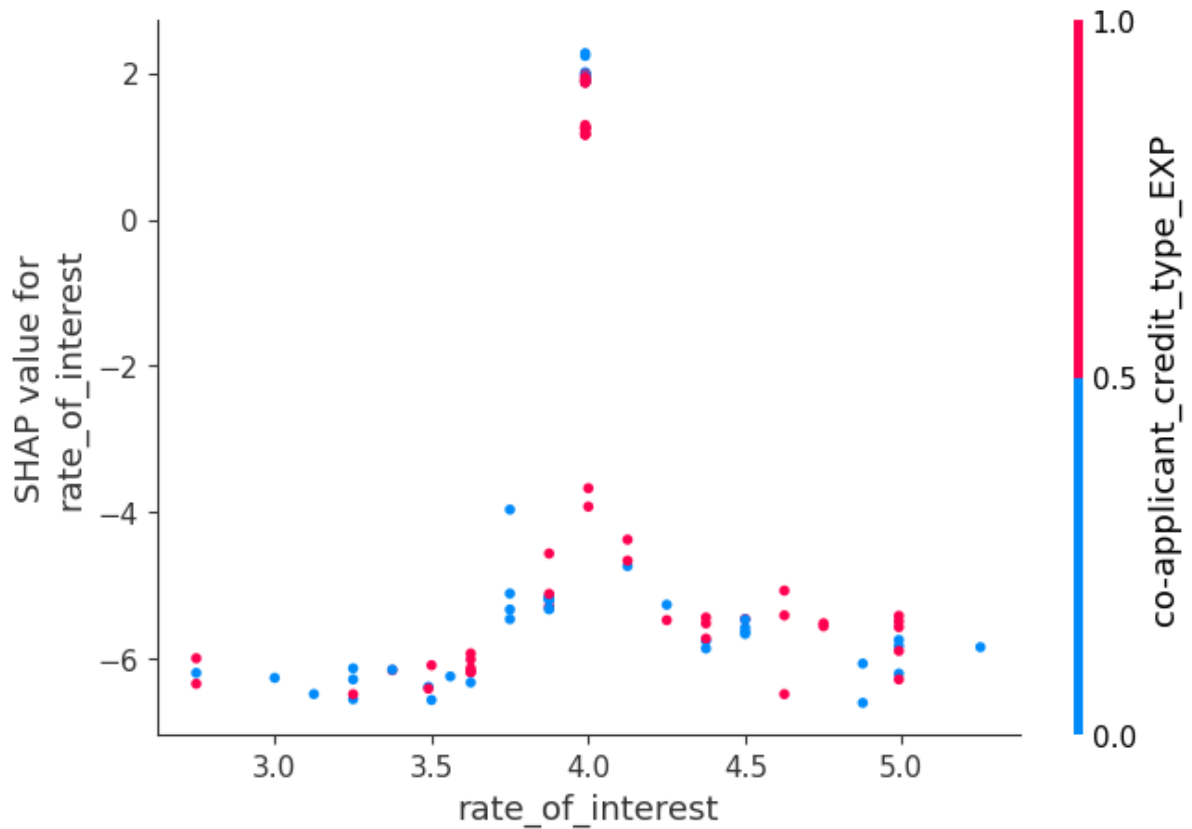


Figure 10 SHAP Dependence Plot for rate_of_interest
Source: Authors

The dependence plot reveals a clear positive relationship between the interest rate and the predicted probability of default. As `rate_of_interest` increases from approximately 2.8% to 5.2%, the SHAP values generally increase, indicating higher default risk. A noticeable non-linear jump is observed around 4.0%, suggesting a potential risk threshold. The coloring by `co-applicant_credit_type_EXP` highlights an interaction effect, implying that the impact of interest rate is moderated by the co-applicant's credit profile.

To understand how the model arrives at a specific prediction, SHAP local explanations were examined using force plots and waterfall plots. The following analysis focuses on a representative test instance (Instance 1000).

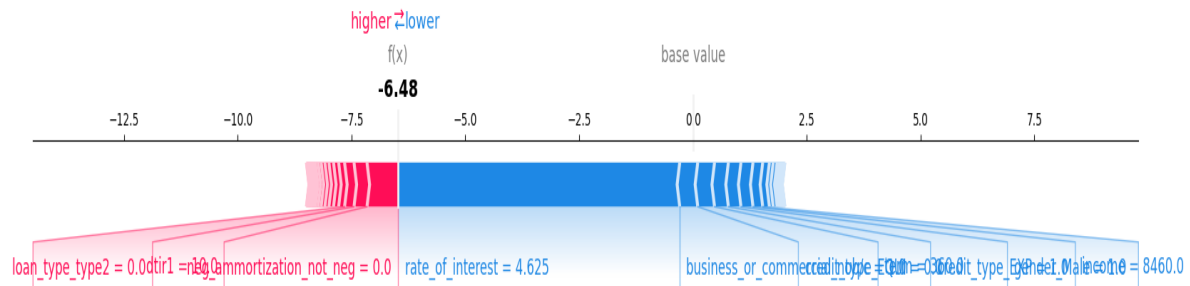


Figure 11 SHAP Force Plot for a Selected Instance
Source: Authors

The SHAP force plot shows how different features influence the prediction for an individual borrower by either increasing or decreasing default risk. Although interest rate increases the likelihood of default, this effect is offset by stronger protective factors such as favourable loan characteristics, lower debt burden and borrower-related factors. Overall, the combined effect of these features leads to a prediction of non-default, showing that the model considers multiple factors rather than relying on a single variable.

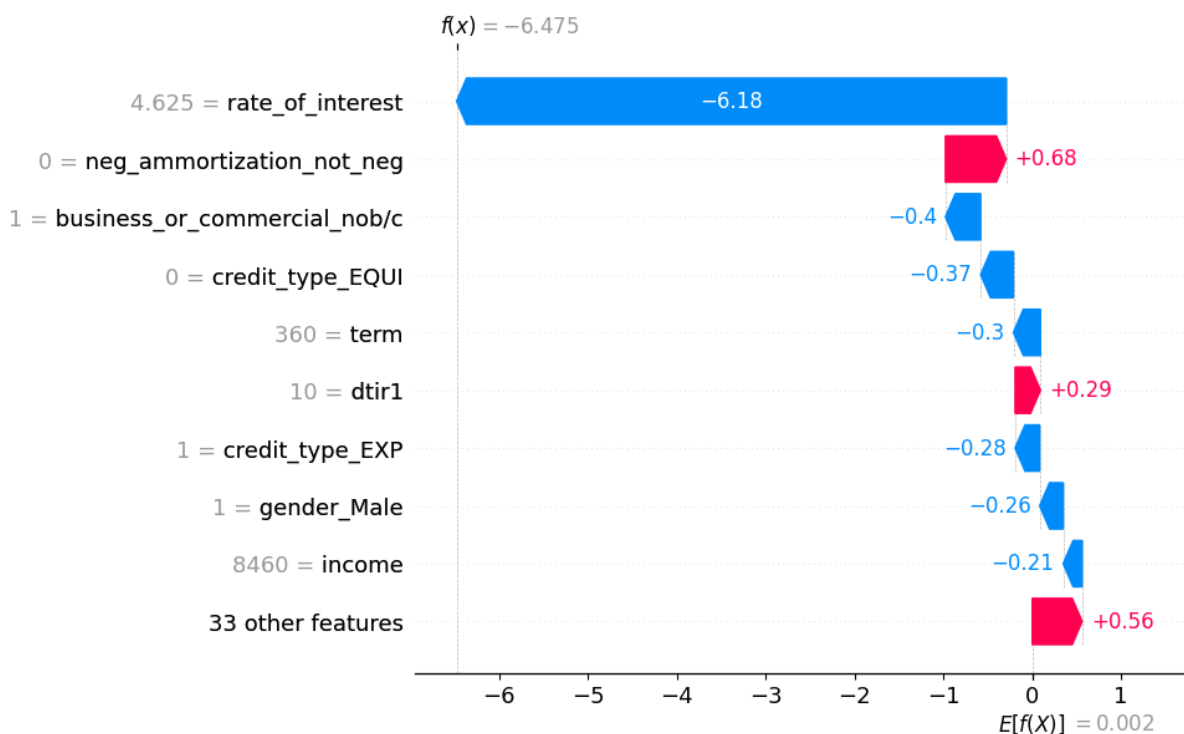


Figure 12 SHAP Waterfall Plot for the Same Instance
Source: Authors

The SHAP waterfall plot provides a step-by-step explanation of how different features contribute to the final prediction for an individual borrower. While some factors increase default risk, other stronger feature contributions to reduce the risk and lead to a non-default prediction. This shows that the model prediction is influenced by a combination of interacting factors rather than a single variable.

Overall, the SHAP analysis shows that important financial features such as interest rate, credit type, LTV and loan characteristics have a strong influence on predictions. Both the global and local explanations suggest that the model captures meaningful risk patterns and provides transparent insights into how predictions are made. This supports the use of the model for more explainable and responsible credit risk assessment.

LIME

LIME (Local Interpretable Model-agnostic Explanations) was used to explain individual predictions. It provides insight into how specific features influence a single prediction, helping to understand why a particular borrower is classified as default or non-default.

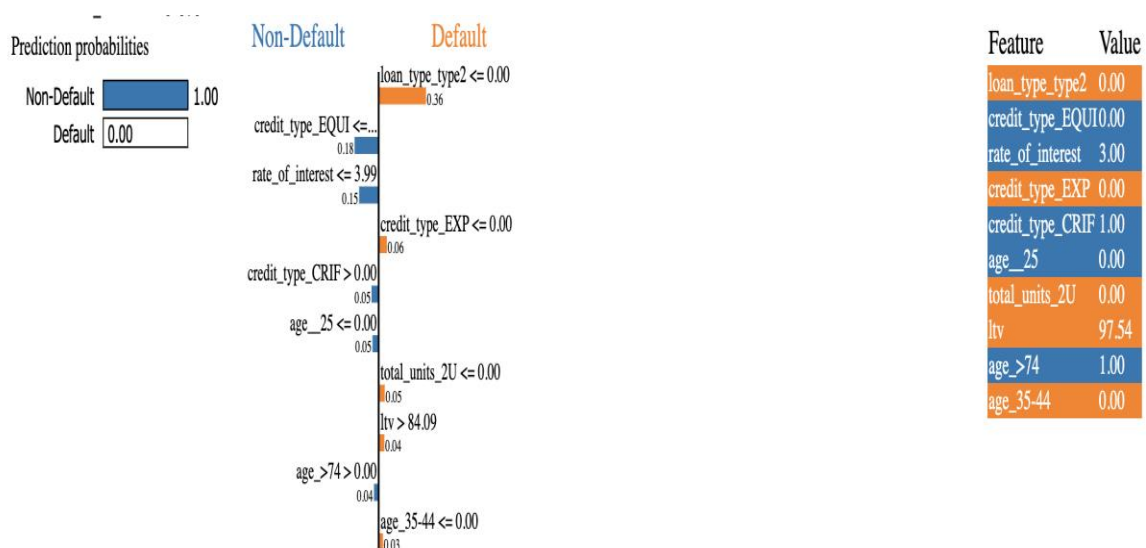


Figure 13: LIME Explanation for Individual Prediction
Source: Authors

LIME was used to explain individual predictions made by the model. Figure 13 shows the explanation for a single prediction. For the selected test instance (index 1000), the model predicts a **non-default outcome with high confidence**, indicating strong certainty in identifying a low-risk borrower. The explanation shows that features such as **credit type and rate of interest** contribute positively towards the non-default prediction, suggesting that stable credit conditions and favourable interest profiles reduce perceived default risk. In contrast, features such as **loan type and higher loan-to-value (LTV)** increase the likelihood of default, reflecting higher financial exposure and risk sensitivity.

The plot shows how different features contribute to this prediction. Features shown in blue support the non-default classification, while features in orange push the prediction towards default. This allows clear and intuitive interpretation of feature influence at an individual level.

However, LIME is based on local perturbation sampling, meaning that explanations are generated by fitting a surrogate linear model around the selected instance. As a result, the method provides an approximation of the model's behaviour rather than an exact representation. This can lead to variability across runs and reduced stability compared to more theoretically grounded approaches.

Overall, LIME provides intuitive, human-readable explanations that improve transparency for individual predictions. However, its reliance on local approximation makes it more suitable for explanation and communication purposes rather than strict model validation or regulatory auditing. When combined with global methods such as SHAP, it contributes to a more complete understanding of model behaviour in loan default prediction systems.

4.7 Comparison between SHAP and LIME

Both SHAP and LIME were applied to the same test instance (index 1000) using the best-performing ensemble model to enable a direct comparison of local explanations. Both methods show strong agreement in identifying **loan type, interest rate level, credit type, and age-related variables** as the most influential features contributing towards the **non-default prediction**, indicating consistent model behavior across different interpretability approaches.

The empirical results are summarized below.

Table 2 Comparison of SHAP and LIME

Metric	LIME	SHAP
Runtime (seconds)	0.1160	0.0602
Stability	0.5514	1.0000
Fidelity	0.5876	~1.0000
Features (80% contribution)	6	7

Source: Authors

SHAP demonstrates perfect stability (1.0), reflecting its deterministic nature and consistent feature attribution. This is further strengthened in tree-based models through **TreeSHAP**,

which computes exact Shapley values efficiently using the structure of ensemble trees. In contrast, LIME shows moderate stability (0.5514) due to reliance on random perturbations during local sampling, leading to variation across runs.

A similar pattern is observed in fidelity, where SHAP achieves near-perfect alignment with the model output due to its additive decomposition, while LIME (0.5876) only approximates the local decision boundary using a surrogate linear model. SHAP is also more computationally efficient in this implementation (0.0602s vs 0.1160s), while LIME produces slightly more compact explanations with fewer features.

Overall, SHAP provides more stable, accurate, and theoretically grounded explanations, making it more suitable for audit and regulatory use, particularly in conjunction with TreeSHAP for tree-based models. LIME offers simpler and more interpretable local explanations but with reduced stability and fidelity. Together, they highlight a clear trade-off between **robustness (SHAP)** and **interpretability simplicity (LIME)** in loan default prediction systems.

Chapter 5: Conclusion

This project focused on developing an accurate and interpretable framework for loan default prediction using machine learning and explainable AI techniques. Different baseline and ensemble models were implemented and compared to identify the most effective approach for credit risk prediction. In addition, SHAP and LIME were applied to improve the interpretability of the final model and provide clearer explanations of predictions.

5.1 Summary of Findings

The results of the study show that ensemble models perform better than baseline models in predicting loan default. Among all models, XGBoost achieved the best overall performance, especially in terms of F1-score and recall, which are important for handling imbalanced data. Random Forest also showed strong performance, while Logistic Regression had the lowest performance, indicating its limitations in capturing complex patterns in the dataset.

The application of SMOTE helped to address the issue of class imbalance and improved the model's ability to detect default cases. In addition, the use of SHAP and LIME provided useful insights into model predictions. SHAP helped identify the most important features affecting loan default, while LIME explained individual predictions. These results show that it is possible to achieve both high accuracy and interpretability in credit risk prediction.

5.2 Achievement of Objectives

The objectives of this project were successfully achieved. Baseline models such as Logistic Regression and Decision Tree were implemented and evaluated, providing a foundation for comparison. Ensemble models, including Random Forest and XGBoost, were developed and showed improved performance compared to baseline models.

Explainable AI techniques, specifically SHAP and LIME, were applied to interpret model predictions. These methods helped identify key features influencing loan default and provided explanations for individual predictions.

The study also analysed the trade-off between accuracy and interpretability, showing that while ensemble models provide better performance, interpretability can be improved using XAI techniques.

Overall, the project successfully developed an interpretable and reliable framework for credit risk prediction.

5.3 Limitations of the Study

Despite the successful outcomes, this study has some limitations. First, the analysis was conducted using a single dataset, which may limit the generalisation of the results to other financial datasets.

Second, while SHAP and LIME were used to provide explanations, the stability and consistency of these explanations were not extensively tested across multiple samples or models.

Also, some preprocessing decisions such as feature removal and imputation methods, may affect the model results.

In addition, advanced deep learning models were not explored in this study, which could potentially improve performance further.

Further, the feature engineering process was limited, and more advanced techniques could further improve model performance. Finally, the study focused on structured financial data, and other data sources such as behavioural or transactional data were not considered.

5.4 Future Work

This study can be extended in several ways in the future. First, the models can be tested on multiple datasets to improve the generalisability of the results across different financial scenarios.

Second, further analysis can be carried out to evaluate the stability and consistency of explanations provided by SHAP and LIME. This would help improve the reliability of explainable AI techniques.

In addition, more advanced models, such as deep learning approaches, can be explored to further improve predictive performance.

Future research can also explore hybrid approaches that combine multiple models to further improve prediction accuracy. In addition, the use of real-time data and deployment of the model in live systems can be investigated to evaluate its practical performance.

5.5 Closing Statement

In conclusion, this study demonstrates that combining machine learning models with explainable AI techniques can provide both accurate and transparent solutions for credit risk prediction. The proposed framework highlights the importance of balancing predictive performance with interpretability, making it suitable for real-world financial applications.

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APPENDIX I

Suman Pokhrel (B01819324)

Self-Reflection

This project gave me a valuable opportunity to work with real-world financial data and understand how machine learning can be applied to credit risk prediction. My main role in the group was to focus on baseline models such as Logistic Regression and Decision Tree, as well as data preprocessing and feature analysis. Through this work, I gained practical experience in applying concepts that I had previously only studied in theory.

One of my main contributions was working on data preprocessing, which I came to understand is one of the most important parts of model development. This included removing irrelevant features, handling missing values, encoding categorical variables and preparing the dataset for modelling. I also carried out basic exploratory data analysis to understand the structure of the data and identify patterns in variables such as income, loan amount and credit-related features. This process helped me realise that the quality of data preparation can strongly influence model performance.

By implementing baseline models in Python using Google Colab, I improved my understanding of how machine learning models are applied in practice. Logistic Regression helped me understand how features relate to the probability of loan default, while Decision Tree showed how predictions can be made using rule-based decisions. Although I did not develop the algorithms from scratch, I learned how to implement them, interpret the outputs and understand their behaviour using the dataset.

I also worked on evaluating model performance using metrics such as accuracy, precision, recall, F1-score and ROC-AUC. This helped me understand that accuracy alone is not enough, especially when dealing with imbalanced data. Learning how to interpret these metrics improved my understanding of how models should be assessed in a practical credit risk problem.

One of the main challenges I faced was deciding which preprocessing techniques were appropriate and understanding how to interpret model outputs. At the beginning, some steps such as feature removal, SMOTE and scaling were difficult to understand, and I had to learn through experimentation and discussion. This helped me develop a more practical problem-solving approach rather than relying only on theory.

Another challenge was ensuring that my work aligned with the work of other group members. Because the project involved different components completed by different members, coordination was important. Through regular discussions, we were able to share ideas, solve problems together and make sure the final project remained consistent. This improved my teamwork and communication skills.

The project also helped me improve my time management. Working to deadlines and completing tasks at different stages of the project taught me how to organise my work more effectively. I became more confident in planning tasks and managing progress over time.

Overall, this project helped me develop practical skills in data preprocessing, model implementation and results interpretation. It also improved my confidence in using machine learning tools and applying them to a real-world financial problem. More importantly, it showed me how technical knowledge, problem solving and teamwork all contribute to completing a successful project.

Critical Appraisal

This project aimed to develop a framework for loan default prediction by combining machine learning models with explainable AI techniques. One of the main strengths of the project is the structured approach followed throughout the study. Starting with baseline models and then moving to advanced ensemble models allowed for a clear comparison of performance and helped show how model complexity can improve prediction results.

The use of baseline models such as Logistic Regression and Decision Tree was an important strength because they provided interpretable benchmarks for comparison. These models helped support understanding of the dataset and key factors influencing loan default. Although they did not achieve the highest performance, they were useful for comparing predictive accuracy with interpretability.

Another strength of the project is the use of multiple performance metrics, including precision, recall, F1-score and ROC-AUC. This provided a more balanced evaluation than relying on accuracy alone, especially given the imbalanced nature of the dataset. The use of SMOTE was also a strength, as it helped improve the model's ability to detect default cases and contributed to better model performance.

The integration of explainable AI techniques such as SHAP and LIME was another positive aspect of the project. These methods improved transparency by helping explain model

predictions and identify important risk factors. This is particularly relevant in financial applications, where trust and accountability are important.

However, the project also has limitations. One limitation is that the analysis was based on a single dataset, which may limit how well the findings apply to other financial contexts. Testing the framework on multiple datasets would improve generalisability and provide stronger evidence of model reliability.

Another limitation is that, although SHAP and LIME were applied, the comparison between these methods was not explored in significant depth. For example, the consistency and stability of explanations across multiple instances were not fully assessed. This means the reliability of the explainability methods could have been evaluated further.

The project also focused mainly on traditional machine learning models, while more advanced approaches such as deep learning were not explored. In addition, feature engineering was limited and could have been developed further to improve model performance.

Another area for improvement is the evaluation process. Although multiple performance metrics, 5-fold cross-validation and a validation set were used, the validation results were mainly used to confirm model stability. Future work could make greater use of repeated validation or additional testing to strengthen the reliability of results.

Reflecting critically, the project achieved its aim of developing an interpretable and reliable framework for credit risk prediction, but it also showed that balancing model performance and interpretability remains challenging. While the chosen methods produced strong results, there are opportunities to improve both predictive performance and the depth of explainability analysis.

Overall, the project demonstrates a good balance between strengths and limitations. It provided a practical framework for credit risk prediction while also highlighting areas for improvement that could be addressed in future work. Through this critical appraisal, I recognise that evaluating both what worked well and what could be improved is an important part of understanding the quality and limitations of a project.

Self-Reflection

Working on the ensemble modelling component of this project was a highly valuable experience that enabled me to move beyond theoretical knowledge and develop practical machine learning skills. Before starting the project, my understanding of models such as Random Forest and XGBoost was mainly based on lectures and academic readings, with limited hands-on application. Through working with a real-world financial dataset, I developed a clearer and deeper understanding of how these models operate and why they are effective in predicting complex outcomes such as loan default.

I learned that Random Forest reduces overfitting by combining multiple decision trees trained on different subsets of the data, while XGBoost improves predictive performance by sequentially correcting errors made by earlier models. Applying these techniques using Python, particularly through libraries such as Scikit-learn and XGBoost, allowed me to gain confidence in building, training, and evaluating machine learning models. I also developed practical experience in hyperparameter tuning using GridSearchCV and cross-validation, which helped me understand how adjusting parameters can significantly influence model performance and stability.

Another important aspect of my learning was data preparation. I worked with scaled datasets and applied SMOTE to address class imbalance, which improved the model's ability to detect loan default cases. This experience highlighted the importance of preprocessing and showed me that model performance is highly dependent on the quality and structure of the data. Without addressing imbalance, a model may appear accurate while failing to identify critical high-risk cases, which would be problematic in real-world financial applications where incorrect predictions can have serious consequences.

One of my key strengths during the project was my ability to critically evaluate model performance. Rather than relying solely on accuracy, I focused on metrics such as recall, F1-score, and AUC-ROC, which provide a more meaningful assessment in imbalanced datasets. This approach allowed me to better understand the strengths and weaknesses of each model. I also contributed to group discussions by interpreting model outputs and communicating findings clearly, which supported the integration of ensemble modelling into the overall project and improved collaboration.

Despite these strengths, I faced several challenges that contributed to my development. Understanding XGBoost was particularly difficult due to its complexity and the large number

of hyperparameters involved. It required independent research, experimentation, and repeated testing to fully understand how different parameters affected performance. Another challenge was balancing predictive accuracy with interpretability. Although ensemble models produced strong results, they operate as black-box systems, making it difficult to explain how decisions are made. This shows the importance of transparency and accountability, especially in financial contexts where decisions can directly affect individuals.

Overall, this experience significantly improved my technical skills, analytical thinking, and problem-solving abilities. It also strengthened my confidence in applying machine learning techniques in real-world scenarios. I developed a more critical perspective on model evaluation, recognizing that strong performance must be balanced with interpretability and ethical considerations therefore playing an important role in preparing me for future roles in data analytics and machine learning, where both technical expertise and critical judgement are essential for responsible decision-making.

Critical Appraisal

From a critical perspective, the project adopted a structured and systematic methodology, which is a key strength. The integration of baseline models with advanced ensemble methods provided a clear framework for comparing interpretability and predictive performance. This approach allowed for a balanced evaluation of different modelling techniques and ensured that the research objectives were effectively addressed in a logical and organized manner.

The selection of Random Forest and XGBoost was appropriate given the complexity of the dataset. Both models demonstrated strong performance, particularly in terms of recall and F1-score, which are crucial in credit risk prediction where identifying default cases is more important than achieving high overall accuracy. Their ability to capture non-linear relationships and interactions between features significantly improved predictive performance compared to simpler models, making them suitable for real-world applications.

However, this improvement comes with a limitation. Both models function as black-box systems, making it difficult to interpret how predictions are generated. This lack of transparency can be problematic in financial decision-making, where organizations are required to explain and justify their decisions to regulators and stakeholders. While the project attempted to address this issue by incorporating explainability techniques such as SHAP and LIME, the evaluation of these methods was limited. In particular, the project did not fully assess the consistency and reliability of these explanations across different datasets or conditions.

The data preprocessing techniques used in the project, including encoding, scaling, and stratified data splitting, were appropriate and contributed to effective model development. The use of SMOTE was particularly beneficial in addressing class imbalance and improving the model's ability to detect default cases. However, the use of synthetic data introduces potential limitations, as it may not fully represent real-world patterns, which could affect the model's generalizability when applied to unseen data.

A key strength of the project was the use of multiple evaluation metrics, including recall, F1-score, and AUC-ROC. This demonstrates a strong understanding of the limitations of accuracy and provides a more comprehensive evaluation of model performance. However, the study has several limitations that should be considered. The use of a single dataset restricts the generalizability of the findings, and limited feature engineering may have prevented the models from reaching their full predictive potential.

In addition, the project did not explore alternative advanced techniques such as deep learning models, which could provide further improvements or insights into the data. The assumption that the dataset fully represents real-world credit risk conditions is also a limitation, as real-world data often includes temporal and behavioral factors that were not captured in this study. These factors could influence model performance in practical applications.

Overall, the project provides a strong and practical framework by combining high-performing ensemble models with explainability techniques. However, for real-world implementation, additional considerations such as bias detection, fairness, regulatory compliance, and ongoing model monitoring would be necessary to ensure that the system is reliable, ethical, and effective in practice.

Sanjiv Shrestha (B01818608)

Self-Reflection

This project provided me with a valuable opportunity to develop my understanding of Explainable Artificial Intelligence (XAI), particularly using SHAP (SHapley Additive exPlanations). My main role was to apply SHAP to machine learning models used for loan default prediction and to analyse how effectively it explains model behaviour. This experience allowed me to connect theoretical knowledge with practical implementation in a real-world financial context.

One of my main strengths during this project was my ability to learn and apply a complex technique independently. At the beginning, my understanding of SHAP was limited, but through research and practice, I was able to successfully implement TreeSHAP with the XGBoost model. I generated different visualisations such as summary plots, feature importance charts, force plots, and waterfall plots. These helped identify key features influencing loan default, such as interest rate, loan characteristics and credit-related variables. This improved my ability to interpret model outputs and understand how different features contribute to predictions in a structured and meaningful way.

Another strength I demonstrated was problem-solving. Working with SHAP on a large dataset required careful handling of computational resources. Since calculating SHAP values for all data points was time-consuming, I used representative samples to make the process more efficient. This helped me understand the importance of optimisation in real-world machine learning tasks, where time and resource constraints are important. I also worked with both global explanations, which show overall feature importance, and local explanations, which explain individual predictions. This improved my understanding of model behaviour at different levels and strengthened my analytical thinking and decision-making approach.

However, I also faced challenges. One key difficulty was moving beyond basic interpretation of SHAP outputs. Initially, I could describe what the visualisations showed, but I found it harder to critically analyse their deeper meaning. Over time, I improved by linking the explanations to model behaviour and questioning their reliability and consistency.

Another challenge was evaluating the stability of SHAP explanations. Since financial data often includes imbalanced classes and correlated features, I needed to check whether similar instances produced similar explanations. This required additional analysis and helped me develop a more critical and research-focused mindset. It also made me realise that

explainability is not only about generating results but also about evaluating their trustworthiness and consistency across different cases.

In terms of skill development, this project improved my Python programming skills, especially in using libraries such as SHAP, pandas, and scikit-learn. I also developed better data analysis and visualisation skills. More importantly, I gained a deeper understanding of the importance of interpretability in machine learning. I realised that accurate predictions alone are not enough, especially in financial applications where decisions must be transparent, explainable, and justifiable.

Overall, this project increased my confidence in working with machine learning and explainability techniques. It also improved my critical thinking and problem-solving abilities, which will be valuable in future roles in data analytics where transparency, accountability, and trust are essential.

Critical Appraisal

The methodology used in this project, particularly for the SHAP explainability component, is well aligned with the aim of developing an interpretable loan default prediction framework. SHAP is an appropriate choice because it is based on a strong theoretical foundation and can provide both global and local explanations. The use of TreeSHAP for tree-based models such as XGBoost is especially effective, as it allows efficient and accurate computation of feature contributions, making it suitable for large-scale financial datasets.

One of the key strengths of the approach is the integration of explainability with high-performing machine learning models. Ensemble models such as Random Forest and XGBoost provide strong predictive performance but are often difficult to interpret. By applying SHAP, the study improves transparency and helps explain how predictions are made. The ability to generate both overall feature importance and individual explanations adds practical value, particularly in financial decision-making where transparency and accountability are required.

Another strength is the effort to evaluate the reliability of explanations. The study does not only present visualisations but also considers the stability of SHAP outputs by analysing similar instances. This is important because explainability methods must produce consistent results to be useful in real-world applications. This step reflects a more advanced and critical approach compared to studies that only focus on visual interpretation.

However, there are several limitations. One major limitation is the use of a single dataset. Although the dataset is suitable for analysis, relying on one dataset reduces the generalisability

of the findings. Different datasets may produce different patterns, and SHAP explanations may vary depending on the characteristics of the data.

Another limitation relates to feature correlation. SHAP assumes that features are independent, but in financial datasets, many variables are correlated. This can affect the accuracy of feature contributions, as the importance of one feature may depend on others. As a result, the explanations may not fully reflect real-world relationships and should be interpreted carefully.

The evaluation of SHAP stability is also somewhat limited. A more detailed analysis could include testing explanations across multiple samples, different models, or alternative preprocessing techniques. This would provide stronger evidence of reliability and improve the robustness of the findings.

In addition, SHAP explains model behaviour rather than real-world causation. The explanations show how features influence predictions within the model, but they do not prove that these features cause loan default. This distinction is important in financial applications, where incorrect interpretation could lead to misleading conclusions.

From a technical perspective, the tools used, such as Python, SHAP, and scikit-learn, are appropriate and widely accepted in both academic and industry contexts. However, computational cost remains a challenge when working with large datasets.

Ethically, the study takes a responsible approach by focusing on interpretability rather than full automation. However, incorporating fairness and bias analysis could further strengthen the study.

In conclusion, the SHAP-based methodology is effective in improving interpretability, but it has limitations related to data dependency, feature correlation, and lack of causal interpretation.

Self-Reflection

This component of the group project provided me with an opportunity to develop a deeper understanding of Explainable Artificial Intelligence (XAI), particularly focusing on local interpretability techniques and their practical application within a financial risk prediction system. My primary responsibility was to implement Local Interpretable Model-agnostic Explanations (LIME) and to conduct a structured comparison between LIME and SHAP to evaluate their effectiveness in explaining loan default predictions.

One of my key strengths during this process was my ability to move from a theoretical understanding of explainability methods to practical implementation. I successfully implemented LIME using Python in Google Colab and integrated it with the trained ensemble model developed by the group. This required a clear understanding of how LIME operates, specifically its use of perturbation-based sampling and local surrogate modelling to approximate complex decision boundaries. Through this process, I developed the ability to interpret how individual features influence model predictions at a local level, particularly in the context of financial variables such as interest rate, credit type, and loan characteristics.

In addition to implementation, I focused on critically evaluating LIME rather than simply presenting its outputs. I designed experiments to assess the reliability of explanations, including analysing the consistency of feature importance across multiple runs and evaluating how well the surrogate model represents the behaviour of the underlying model. This helped me move beyond descriptive interpretation and towards a more analytical approach, where I considered not only what the explanations show, but also how trustworthy they are.

A major challenge I encountered was the inherent instability of LIME explanations. Because the method relies on random perturbation sampling, repeated explanations for the same instance produced variations in feature attributions. This required careful interpretation and highlighted the limitations of relying solely on LIME in high-stakes applications such as credit risk assessment. I also found that parameter selection, including the number of samples and feature selection settings, had a significant impact on the quality of explanations, requiring iterative experimentation and validation.

Working alongside SHAP outputs implemented by my group member further enhanced my understanding of different explainability approaches. While LIME provided intuitive and easily interpretable explanations, SHAP offered more consistent and theoretically grounded

results. This comparison allowed me to critically assess the trade-offs between interpretability, stability, and fidelity, rather than treating explainability methods as interchangeable tools.

From a skills development perspective, this project significantly improved my ability to implement XAI techniques using Python, work with machine learning models in a structured pipeline, and critically evaluate model explanations. It also strengthened my awareness of the regulatory and ethical considerations associated with AI in finance, particularly the importance of transparency, consistency, and accountability in automated decision-making systems.

Overall, this experience reinforced my interest in responsible AI and improved my confidence in bridging the gap between complex machine learning models and interpretable, user-friendly insights.

Critical Appraisal

The methodology adopted for the implementation of LIME and its comparison with SHAP was appropriate for the objectives of the project, particularly in the context of enhancing interpretability in a loan default prediction system. LIME was a suitable choice due to its model-agnostic nature, allowing it to be applied to the ensemble model without requiring access to internal model structures. This flexibility makes it a practical tool for post-hoc interpretability across a wide range of machine learning models.

The application of LIME provided valuable insights into individual predictions, offering clear and intuitive explanations that are accessible to non-technical stakeholders. Its ability to highlight the contribution of specific features at the instance level supports transparency and helps in understanding model behaviour in real-world decision-making scenarios. This is particularly relevant in financial applications, where explanations must often be communicated to customers or business users.

However, the evaluation also revealed several limitations of LIME. One of the most significant issues is its reliance on local surrogate modelling, which approximates the behaviour of the underlying model rather than representing it exactly. In cases where the model exhibits complex non-linear relationships, the linear approximation used by LIME may fail to capture important feature interactions, reducing the accuracy of explanations.

Another critical limitation is the instability of LIME explanations. Due to its dependence on stochastic perturbation sampling, explanations can vary across runs, even for the same input instance. While the most influential features tend to remain consistent, variation in lower-ranked features can affect the overall interpretation. In regulated environments such as credit

risk assessment, where explanations must be consistent and reproducible, this variability presents a significant challenge.

In comparison, SHAP demonstrated clear advantages in terms of both stability and theoretical soundness. Its foundation in cooperative game theory ensures properties such as additivity and consistency, resulting in explanations that are both reliable and mathematically grounded. The deterministic nature of SHAP further ensures that repeated evaluations produce identical results, which is a critical requirement in high-stakes applications.

Despite these limitations, LIME retains practical value as a complementary explainability technique. Its explanations are more intuitive and easier to interpret, making it particularly useful for communication and user-facing applications. This highlights an important trade-off between robustness and interpretability: while SHAP provides more accurate and stable explanations, LIME offers greater simplicity and accessibility.

There are also broader limitations to this analysis. The evaluation was conducted within a specific dataset and modelling framework, which may limit the generalisability of the findings. Additionally, the effectiveness of LIME is sensitive to parameter choices, and different configurations may lead to variations in explanation quality. Further work could explore methods to improve stability, such as controlled sampling strategies or hybrid approaches that combine multiple explanation techniques.

In conclusion, this component of the project demonstrated that LIME and SHAP serve complementary roles in explainable AI. SHAP is more suitable for robust and reliable analysis, particularly in regulatory contexts, while LIME provides intuitive explanations that enhance user understanding. The comparative evaluation contributes to a more balanced perspective on explainability, emphasising that the choice of method should depend on the specific requirements of the application rather than assuming a single universally optimal approach.

APPENDIX II

1. Link to the dataset

Raw Dataset

https://studentmailuwsac-my.sharepoint.com/:x:/g/personal/b01818608_studentmail_uws_ac_uk/IQC2h3gC3VRpTa4LyNkVSQtDAXwBeQBJDCAtumCEuZE-YWc?e=PmjLxV

Cleaned Dataset

https://studentmailuwsac-my.sharepoint.com/:x:/g/personal/b01818608_studentmail_uws_ac_uk/IQDvn_v9nCTJgDNwigkDxEAYFr4NDud7Tk3cCGZJeioY?e=qoJ35X

2. Project Process Documentation

MSc Masters Project

PROJECT PROCESS DOCUMENTATION

Student(s): Suman Pokhrel, Abhinaw Adhikari, Sanjiv Shrestha, Alish Dahal
Supervisor: Dr Yan Ge

Meeting Number: 1 **Date/Time:** March 11, 2026 / 1:20 P.M (Duration 30 mins)

Agenda for meeting: *(Should be set and sent to the supervisor before the meeting)*

Project Title and work distribution

Guidance about specification

Objectives and Literature Review

Discussion of agenda items:

- Project Title and work distribution of the report.
- Format of Specification and what to mention as objectives.
- Types of Literatures to include.

Summary of agreed action plan:

- Title of the project is fine.

- Divide the XAI component of SHAP and LIME to two members.
- Follow the provided handbook for format and literatures.

Meeting Number: 2 **Date/Time:** March 26, 2026 / 10:50 A.M (Duration 20 mins)

Agenda for meeting: *(Should be set and sent to the supervisor before the meeting)*

About research gap

Formulating research questions and objectives better

Hypothesis formulation

Discussion of agenda items:

- How to Connect between each component of the overall project?
- How to find the research gap?
- How to formulate the research questions and objectives?
- Is there a need to do hypothesis testing or not?

Summary of agreed action plan:

- Do the overall report in a Series with one model after another.
- Read the literatures to find the limitations and gaps in the existing knowledge.
- The solutions to fill the research gaps should be presented as objectives.
- Do hypothesis testing only if there will be statistical testing.

Meeting Number: 3 **Date/Time:** April 15, 2026 / 10:50 A.M (Duration 20 mins)

Agenda for meeting

Suggestions on EDA and Pre-processing

Discussion about implementation and findings

Idea about models and XAI components.

Discussion of agenda items:

- Do we need to show EDA and Pre-processing in detail?
- How do we show the implementation of the codes?
- Do we need to individually explain the ML models and XAI?
- How to integrate all 4 components of the project in the report?
- How to present the findings?

Summary of agreed action plan:

- Show EDA and pre-processing in implementation.
- Explain the models and critically define them.
- Link the findings to the objectives.

Meeting Number: 4 **Date/Time:** April 20, 2026 / 9:50 A.M(Duration 15 mins)

Agenda for meeting:

About Feedback of specifications.

Ideas about individual sections

Ideas about showing code implementation.

Discussion of agenda items:

- How to update the final project as per the feedback of specifications?
- What to mention in the individual sections?
- Where do we show the codes implemented?

Summary of agreed action plan:

- Make changes as per the feedback received including necessary change in the Title of the Project.
- Self-Reflection and Critical Appraisal as individual sections of the report.
- Save the codes in public repository and share the link.

3. Specifications

Suman Pokhrel (B01819324)

University of the West of Scotland

School of Computing, Engineering and Physical Sciences

MSc Masters Project Specification

Student's Name: Suman Pokhrel

Banner id: B01819324

MSc Programme: MSc Information Technology with Data Analytics

MSc Programme Leader: Dr Volkan Tunali

Project Title

(5 Marks)

Explainable AI for Loan Default Prediction Using Large-Scale Financial Data.
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Abstract

In recent years, financial institutions have increasingly adopted machine learning models to improve the accuracy of credit risk assessment. This group project proposes the development of an Explainable Artificial Intelligence (XAI) framework for loan default prediction using large-scale financial data. Baseline models are now often seen as classical benchmark to evaluate the performance of ensemble models. Advanced algorithms such as Random Forest and gradient boosting methods often outperform traditional statistical approaches like Logistic Regression. However, despite their strong predictive performance, these complex models are often considered ‘black box’ systems, making it difficult to understand and justify their decisions. In highly regulated financial environments, transparency, fairness and accountability are essential.

To address this challenge, the project is structured into four interrelated components. The first component will focus on developing baseline credit risk models and conducting feature-level analysis to understand the key factors influencing loan default. The second component will design and evaluate advanced ensemble machine learning models and compares their predictive performance with traditional baseline models using evaluation metrics. The third component will apply explainability technique of SHAP to interpret complex machine learning models and generate both global and local explanations. The fourth component will conduct LIME explainability technique and a systematic empirical comparison of SHAP and LIME, assessing their stability, fidelity, robustness and suitability in context of the prevailing financial regulations.

The project will use a publicly available, anonymized loan default dataset containing both numerical and categorical variables. Standard data preprocessing, stratified data splitting, and multiple performance metrics such as Accuracy, Precision, Recall, F1-score and ROC-AUC will be employed to ensure fair and reliable evaluation.

Overall, this proposal aims to develop a system that not only predicts loan defaults accurately but also clearly explains how and why those predictions are made using XAI methods and not only relying on ‘black box’ models.

(Rationale for the research showing gaps in the current knowledge and general statements on intent and direction of the project)*

Justification:

Increasing complexity of determining credit risk has encouraged institutions to explore advanced solutions such in AI (artificial intelligence) and ML (machine learning) to improve prediction accuracy and strengthen risk management practices (Ereiz, 2019). Predicting credit default is important for lenders, as poor lending decisions can lead to substantial financial losses. This results to effective and reliable credit risk assessment being a key priority in the financial sector.

Credit risk assessment has relied on statistical techniques like Decision Trees and Logistic Regression. These models have been widely adopted in both academic research and real-world banking systems due to their simplicity, transparency and regulatory acceptance (Aslam, 2019). One of the main advantages of these credit scoring methods is their interpretability, as they allow lenders to clearly understand how individual borrower characteristics influence default risk. For this reason, such models continue to serve as standard baseline approaches in credit risk modelling.

Literatures indicate that advanced predictive models can often gain high predictive precision while comparing to baseline methods. However, their performance must be carefully evaluated against baseline models to demonstrate meaningful improvement and practical usefulness (Shoaeinaeini et al., 2024). Without reliable baseline models, it becomes difficult to assess if the increased complexity of advanced models is justified or not. Therefore, baseline models play an important role in establishing reference performance levels and ensuring transparency in credit risk prediction.

Prior research also highlights the importance of feature analysis in understanding credit default causes. Financial and credit-related features such as income level, loan amount and credit history are consistently identified as influential factors of loan default (Shoaeinaeini et al., 2024). Features such as credit history, income, debt and employment information are important in predicting the credit defaults (Peussa, 2016). However, many studies mainly focus on improving prediction accuracy, with limited attention given to explaining how individual features influence default outcomes.

This lack of attention can reduce the transparency and understanding of credit risk models. Therefore, a more structured analysis of influential features is needed to improve explainability and support clearer credit risk decision-making. This study addresses this gap by focusing on feature analysis alongside baseline models.

This project is justified by the need to establish reliable baseline credit risk models and conduct systematic feature analysis to better understand the factors influencing loan default. By focusing on interpretable baseline models and key features, this study aims to support more transparent and explainable credit risk decision-making and provide a benchmark for evaluating more advanced models within the wider group project.

Overall Aim:

This study is proposed with the aim of developing and evaluating interpretable baseline credit risk models, specifically ‘Logistic Regression’ and ‘Decision Tree’ for loan default prediction and to perform feature-level analysis to identify the most influential financial variables affecting loan default outcomes.

(List of tasks to be undertaken to achieve overall aim of the project and to answer the research question posed)*

The objectives focus on baseline credit risk modelling and feature analysis and define the key tasks required to achieve the aim of this individual study within the group project.

1. To review relevant academic literature on traditional baseline credit risk models, with a focus on commonly used methods. This objective is with aim of establishing a theoretical foundation by understanding how these models have been applied in loan default prediction and why they remain widely used in financial decision-making.
2. To explore and prepare the selected loan default dataset by conducting initial data understanding and descriptive analysis. This includes examining key borrower and financial features, such as income level, loan amount, credit type, security type and repayment behaviour, to understand data distributions, patterns and their potential relationship with loan default outcomes.
3. To develop baseline credit risk prediction models such as logistic regression and decision tree. These models will be implemented to establish reference performance levels for loan default prediction, providing a benchmark against which more complex models will be compared.
4. To perform feature-level analysis to identify important factors influencing loan default, using interpretable model outputs such as regression coefficients and feature importance measures. This objective focuses on understanding how individual features contribute to default and non-default predictions, rather than maximising predictive accuracy.
5. To assess the explainability and transparency of baseline models, by evaluating how effectively feature analysis supports clear interpretation of model decisions. This objective aims to demonstrate the practical value of baseline models in supporting responsible and understandable credit risk assessment.
6. To determine how baseline models perform using classification metrics such as ROC-AUC, confusion matrices, precision, fl-score, accuracy and recall to assess their predictive capability and reliability in loan default classification.

(An account of the current thinking/published information in the specific area of project)*

Credit risk assessment is an important part of financial sector because wrong lending decisions can lead to loan defaults and financial losses. In the past, banks were dependent on manual judgment and experiences of individuals to approve loans. With the increasing availability of large financial datasets, statistical and machine learning models are now widely used to predict whether a borrowing will be default or not. Peussa (2016) in his study titled **Credit Risk Scorecard Estimation by Logistic Regression** applied 'logistic regression' in-order to develop credit scorecards using real consumer data. It revealed that statistical scorecards perform much better than traditional decision-making based only on manager experience. Important features included credit history, income, debt and employment information. This study explains why logistic regression is popular in the financial sector due to its simple, reliable and easy to understand nature.

Ahmed (2019) also discussed traditional credit scoring methods and the use of scorecards in banks in his research. The study explained key risk measures such as Probability of Default and Expected Loss. It showed that structured scorecards help banks make better loan approval decisions compared to manual evaluation, but the paper mainly focused on basic features and traditional approaches, which may not fully capture complex customer behaviour.

Similarly, Aslam (2019) in his research reviewed different loan default prediction studies and found that machine learning models, especially neural networks, usually perform better than logistic regression. The study also highlighted that historical data quality and feature selection are very important for good predictions. Even so, it was noted that traditional models are still useful because their results are easier to interpret.

Another study on similar research topic by Ereiz (2019) tested different machine learning models using microcredit loan data and compared them with simple linear scoring methods. The results showed that advanced models achieved much higher accuracy. The study also highlighted that better results need better quality data and feature engineering has an important role in improving model performance, not just the choice of algorithm.

Anand, Velu, and Whig (2022) in their study used data from multiple internet sources to make predictions and presented a variety of baseline models as well as ensemble models. Their results showed that ensemble models performed better in predicting loan defaults. Various evaluation measures were calculated and was found that among all the models such as Extra Trees Classifier and Random Forest gave the highest Accuracy.

Chen (2022) explained that a customer's credit history is an important factor in deciding the loan amount and interest rate. Because of this, quickly identifying and analysing customer information has become an important research topic. The study applied a penalized linear regression model and a neural network to make bankruptcy predictions more accurate. The best regression method was chosen based on the data and tested to help banks manage risk better. The results showed that the improved models performed better in predicting bankruptcy compared to the other four models.

Recent studies have also highlighted the importance of explainability in predicting credit defaults. Serengil et al. (2022) examined different machine learning methods to predict non-performing loans. The study also used explainable artificial intelligence techniques to better understand how the prediction models make decisions. Several measures, including Precision, Recall, F1 Score, Imbalance Accuracy (IAM) and Specificity were used to check how well the models performed. Algorithms' performances were evaluated and compared along with several explainable AI tool's explainability. The findings indicated that LightGBM achieved the best performance compared with other models.

Shoaeinaeini et al. (2024) used dataset from online source of a finance company. The data was first cleaned, changed into a suitable format, and important features were selected. Several machine learning methods were then tested on the dataset. Among them, the AdaBoost classifier showed the highest accuracy of 88%. The study also compared the improved model with the original version using ROC curve analysis and feature importance. The results showed that the proposed approach performed better, with a higher ROC area and fewer false positive results. In addition, the model depended more on financial information than on personal customer details, showing that it was effective in classifying the data.

Research Gap

Baseline models are important for transparency and benchmarking, yet they are often used only for comparison rather than deeper analysis of their predictive behavior. In addition, feature analysis is frequently treated as a secondary step, resulting in limited interpretability of how financial variables influence credit risk predictions. Therefore, this study focuses on baseline model evaluation and feature analysis to provide clearer, interpretable insights for credit risk prediction.

(Explanation and justification of approach and methodology proposed for the project. This section should explain the nature of the data and method of data collection. Identification and brief discussion of any analytical tools to be used; Identification of any ethical and practical issues.)*

Research Approach

The proposed research will adopt a quantitative design approach using ML methods to analyse financial data and predict loan default. Quantitative approach is suitable for this study because it focuses on numerical data and statistical analysis.

The problem for analysis will be formulated as a binary classification task, as the target variable has two possible outcomes of whether the credit will be default or non-default. According to Chen (2022), binary classification in credit risk and prediction loan default is prevailing. This project will apply supervised machine learning, as the dataset includes labelled outcomes that indicate whether each loan has defaulted.

Nature and Source of Data

The dataset to be used in this project is obtained from a secondary data source named Kaggle with file named 'Loan Default Dataset' (M Yasser H, 2022). The dataset contains financial and demographic information related to customer loans, that makes it suitable for risk and credit default prediction analysis.

As for the nature of data, the dataset consists of a mixture of categorical and numerical features. There are 21 categorical and 12 numerical variables, excluding the target variable. The target variable indicates whether a loan has defaulted or not.

Data Preprocessing

Several preprocessing steps will be applied before model development. Firstly, data cleaning will be performed to identify and handle missing, inconsistent or invalid values. Missing values may be treated using appropriate techniques depending on the nature of the feature. After that, outlier detection and treatment will be carried out to reduce the impact of extreme values that could distort the learning process.

Next, categorical variables will be encoded into numerical forms to process them. Encoding techniques like label encoding or one-hot encoding will be used based on the characteristics of the variables. Finally, normalization will be applied to the numerical data where needed as to keep all the features on similar scale.

Data Splitting

To fairly evaluate the machine learning models, the dataset will be split into training and testing parts. In this study, 70% of the data will be used for training and the remaining 30% for testing their performance.

Stratified sampling will be applied during the splitting process to confirm that both the sets contain a similar ratio of non-default and default cases and to prevent biased model evaluation. A separate test set will help to check if the model works well on data it hasn't seen before.

Analytical Methods

Logistic regression helps measure how different variables influence the likelihood of an event occurring (Sperandei, 2014). This study will apply machine learning classification techniques to predict loan default. Logistic regression will serve as a baseline model for its simplicity and common use in financial risk studies. It provides a clear understanding of how individual features influence the probability of default.

A decision tree is a model that combines a series of simple tests in an organized way, where each numerical variable compares to a limit at each step (Charbuty and Abdulazeez, 2021). Decision tree will be included as it captures complex relationships and is easy to interpret for loan decisions.

Model Evaluation

Model performance will be examined through various evaluation metrics to verify a fair and inclusive comparison.

Accuracy measures the percentage of correct predictions made by a classification model (Shoaeinaeini et al., 2024). Accuracy will measure the overall proportion of correct predictions but may be misleading if the dataset is imbalanced. So, additional metrics will be used.

Precision is the ratio of correctly predicted positive cases to all cases predicted as positive. It shows how well the model identifies positive instances (Shoaeinaeini et al., 2024). So, it will evaluate how many predicted defaults are actual defaults.

Recall, also called the True Positive Rate, measures how many of the actual positive cases the model correctly identifies. It shows how well the model finds all positive outcomes (Shoaeinaeini et al., 2024). Recall will measure well the model identifies actual default cases.

Precision and recall are combined into the F1-score, which is the average of precision and recall (Anand, Velu and Whig, 2022). F1-score will also be used as an evaluation metric.

The ROC curve displays the balance between true positives and false positives, and a higher AUC means the model can distinguish between classes more accurately (Shoaeinaeini et al.,

2024). The ROC-AUC metric will be used to check how well the model can tell apart default and non-default cases at different thresholds. A confusion matrix will be included to provide additional insight into classification performance by analysing true and false prediction outcomes.

Feature Analysis

Feature analysis will be conducted using statistical methods. Correlation analysis will be used to examine relationships between variables and detect highly related features. Logistic regression coefficients will be analysed to understand how each feature influences loan default prediction. Feature selection methods will help identify the most important predictive variables.

Tools and Software

The project will use Python as it is well-suited for data analysis and machine learning. Pandas and NumPy will help with handling and organizing data. Scikit-learn will be used to build and test the models, while Matplotlib and Seaborn will be used to create graphs and visualizations.

Ethical Considerations

Ethical issues will be considered throughout the project. The dataset does not contain any personal identifiers, ensuring privacy and data protection. There is no direct involvement of human participants. Explainability techniques are applied to reduce biased or unfair decision-making and to promote responsible use of machine learning in financial contexts.

(A timetable for completion of the project. MSc Masters Project full-time is allowed 15 weeks and part-time is 30 weeks. You may use any appropriate method to present the well-thought-through plan of action reflecting your project* activities).*

Weeks 1–2:

Review research papers related to loan default prediction, focusing mainly on logistic regression and other basic models to understand how previous studies approach the problem, what features are commonly used, and how the results are interpreted and explained. This will provide a basic understanding of the methods used in this area and help guide the approach for the project.

Weeks 3–5:

Prepare the dataset for analysis including various pre-processing steps. Categorical variables will be encoded where necessary. Exploratory data analysis (EDA) will also be carried out to understand the structure of the data, check distributions, and identify important features that may influence loan default.

Weeks 6–8:

Build baseline models to predict loan default after splitting dataset. Logistic regression will be used as the main baseline model, and other simple models may also be tested. The evaluation methods will be discussed and agreed upon with other members of the group to ensure consistency.

Weeks 9–11:

Use evaluation metrics to determine the performance standard. These metrics will help measure how well the models predict loan default. In addition, feature importance or coefficient analysis will be used to identify which variables have the most influence on the predictions.

Weeks 12–13:

Interpret and discuss the results from the baseline models. The findings from the feature analysis will also be explained to understand which factors are most related to loan default and how they affect the predictions.

Weeks 14–15:

Combine and share the findings with the rest of the group. All sections will be integrated into the final report, and the document will be reviewed and edited before submission.

References

(Correctly presented in *Cite Them Right Harvard* style)

(10 Marks)

Reference list

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**Individual contribution for group project*

Marking scheme

(Marking scheme for final report/dissertation. Note: For group project, each student in a team must submit their own sections for- Presentation of results, Conclusions and Critical Self-Evaluation)

Introduction	5
Literature Review	15
Research Design	15
Implementation	25
Testing	10
Presentation of Results	10
Conclusions	10
Critical Self-Evaluation	10
Total	100

Supervisor

Dr. Yan Ge

Moderator

Dr. Fadi Safieddine

Programme Leader

Dr. Volkan Tunali

Date specification submitted

16 March 2026

Please complete the 'ethics' & pathway confirmation form below for all projects.

School of Computing, Engineering and Physical Sciences

MSc Masters PROJECT – REQUIREMENT FOR ETHICAL APPROVAL & PATHWAY CONFIRMATION

SECTION 1: TO BE COMPLETED BY THE STUDENT(S)

Does your proposed research involve: research with human subjects (including requirements gathering and product/software testing), access to company documents/records, questionnaires, surveys, focus groups and/or other interview techniques? Does your research entail any process which requires ethical approval? (please enter ✓ in the appropriate box)

YES		You must submit an application for approval to the Ethics Review Manager
NO	✓	You do not need to submit an application to the Ethics Review Manager

☒ I confirm that the above project specification aligns with my MSc programme specialist pathway. (please enter ✓ in the box)

Name of student (print name): Suman Pokhrel

Signature: *Suman Pokhrel*

SECTION 2: TO BE COMPLETED BY THE PROJECT SUPERVISOR

I understand that the above project requires/does not require* ethical approval (*please delete as appropriate).

☒ I confirm that the above project aligns with the MSc programme specialist pathway

Student(s) is (are) enrolled in. (please enter ✓ in the box)

Supervisor (print name):

Yan Ge

Signature:

Yan Ge

Date: 15/March/2026

IMPORTANT: please note that by signing this form all signatories are confirming that any potential ethical issues have been considered and, where necessary, an application for ethical approval has been/will be made via the Ethical Review Manager software.

Any project requiring ethical approval but which has not been given approval will not be accepted for marking.

Ethical approval cannot be sought in retrospect.

